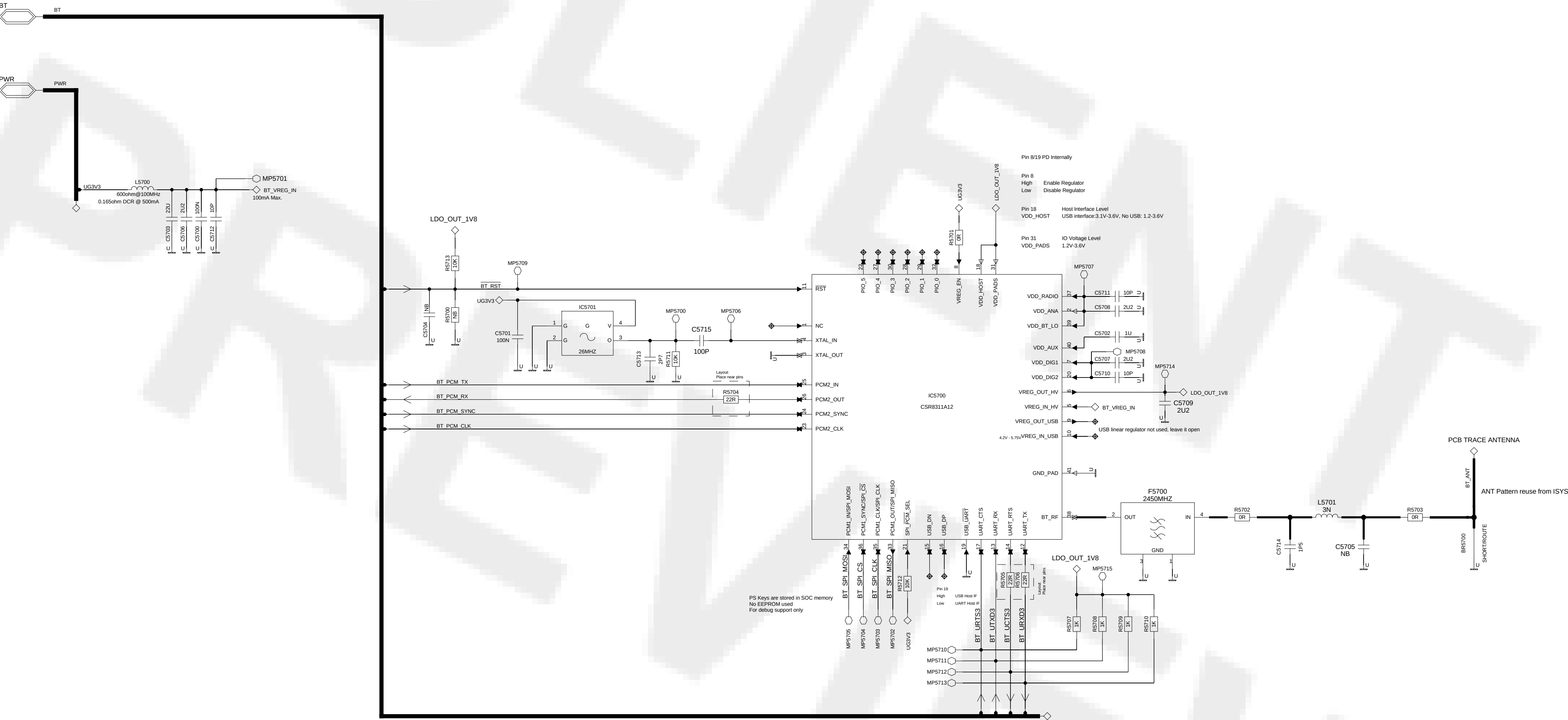


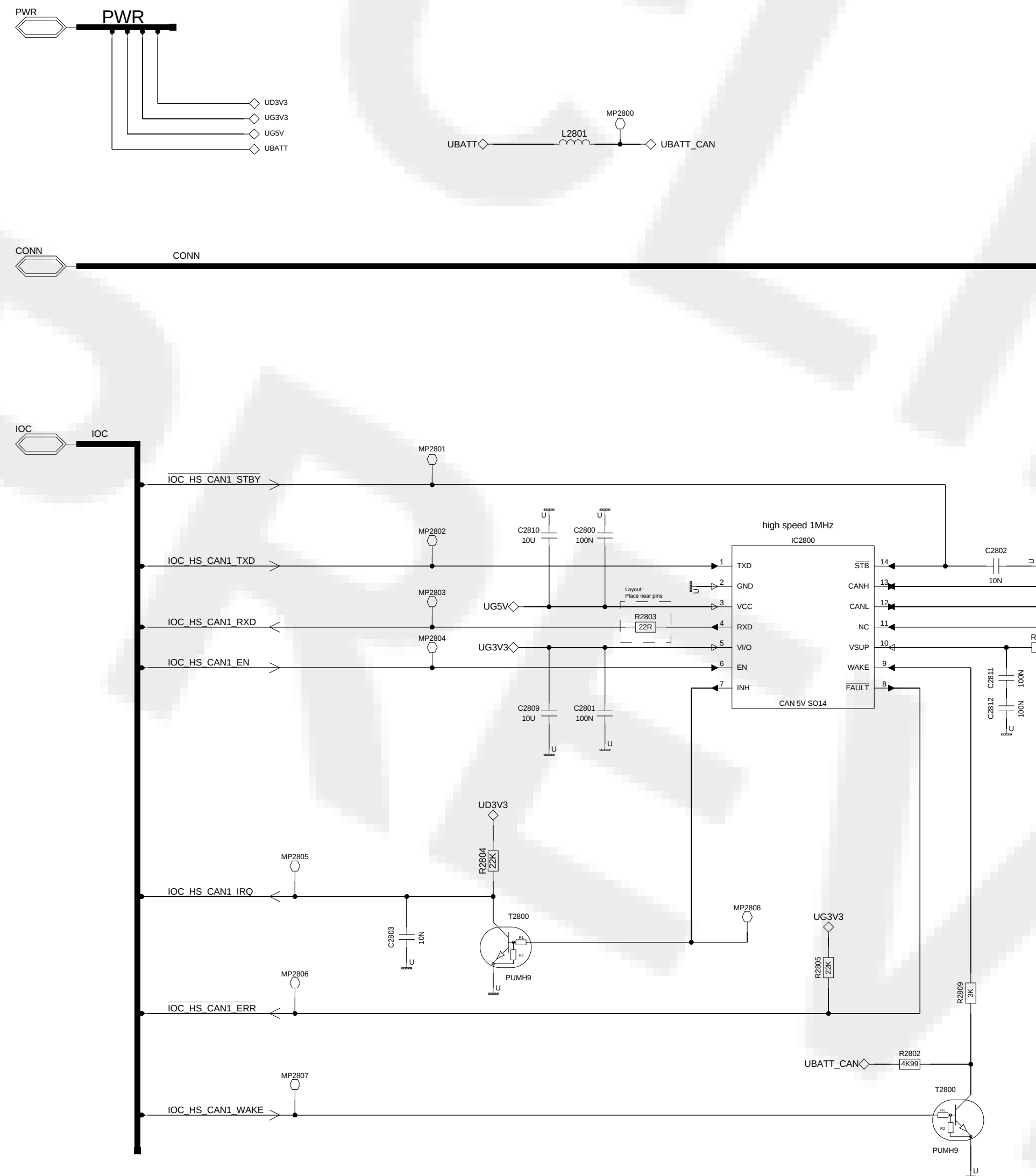
BLUETOOTH (5700-5799)



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Rev.	type of change		change-no.	date of change	auth. checked	
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			created			
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4010_Bluetooth			2	31		
			Type	Variant	Part Number	
substitute for:			substitute by:	C143		

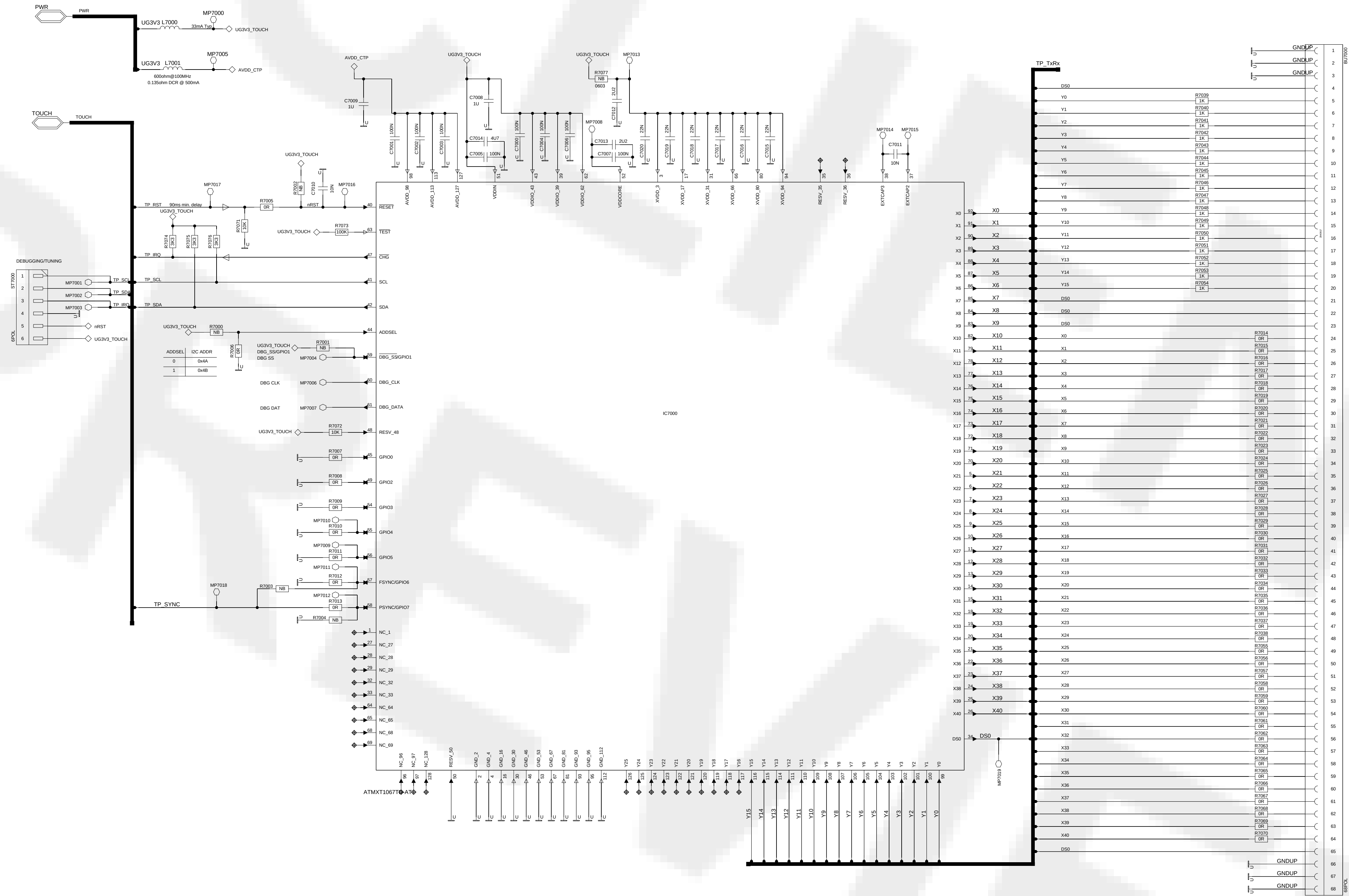
CAN (2800-2899)



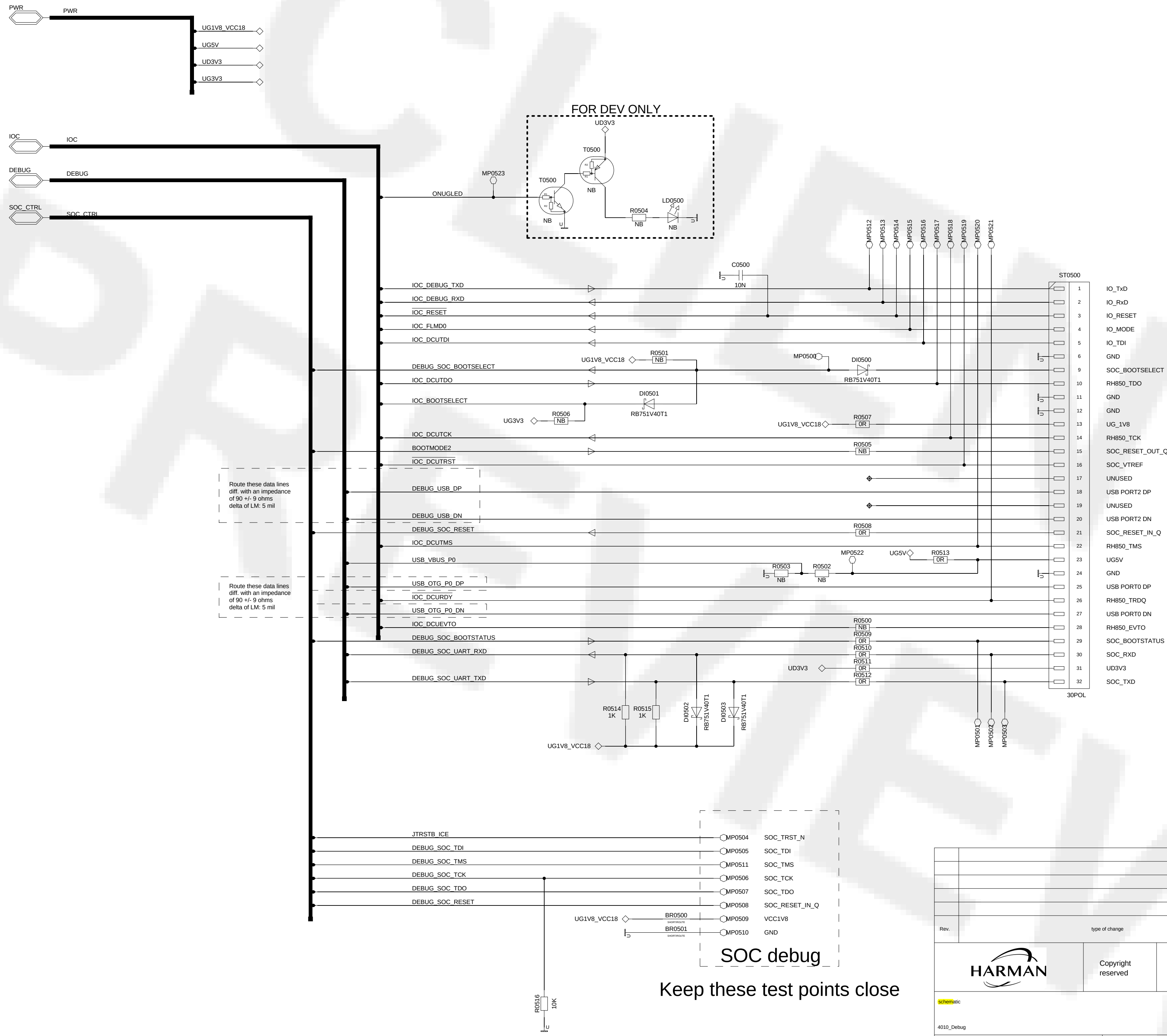
Layout:
place close to
car connector

Rev.	type of change					change-no.	date of change		auth.	checked
			Copyright reserved	Wiring modifications reserved	date			name		
					created					
					checked					
 schematic						Sheet	Sheets of		Sample	
						3	31			
4010_CAN						Type	Variant		Part Number	
substitute for:				substitute by:						

CTP (7000-7099)



Rev.	type of change	change no.	date of change	auth.	checked
			date	name	
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Sheet 4	Sheets of 31	Sample			
Type	Variant	Part Number			
substitute for:	substitute by:				



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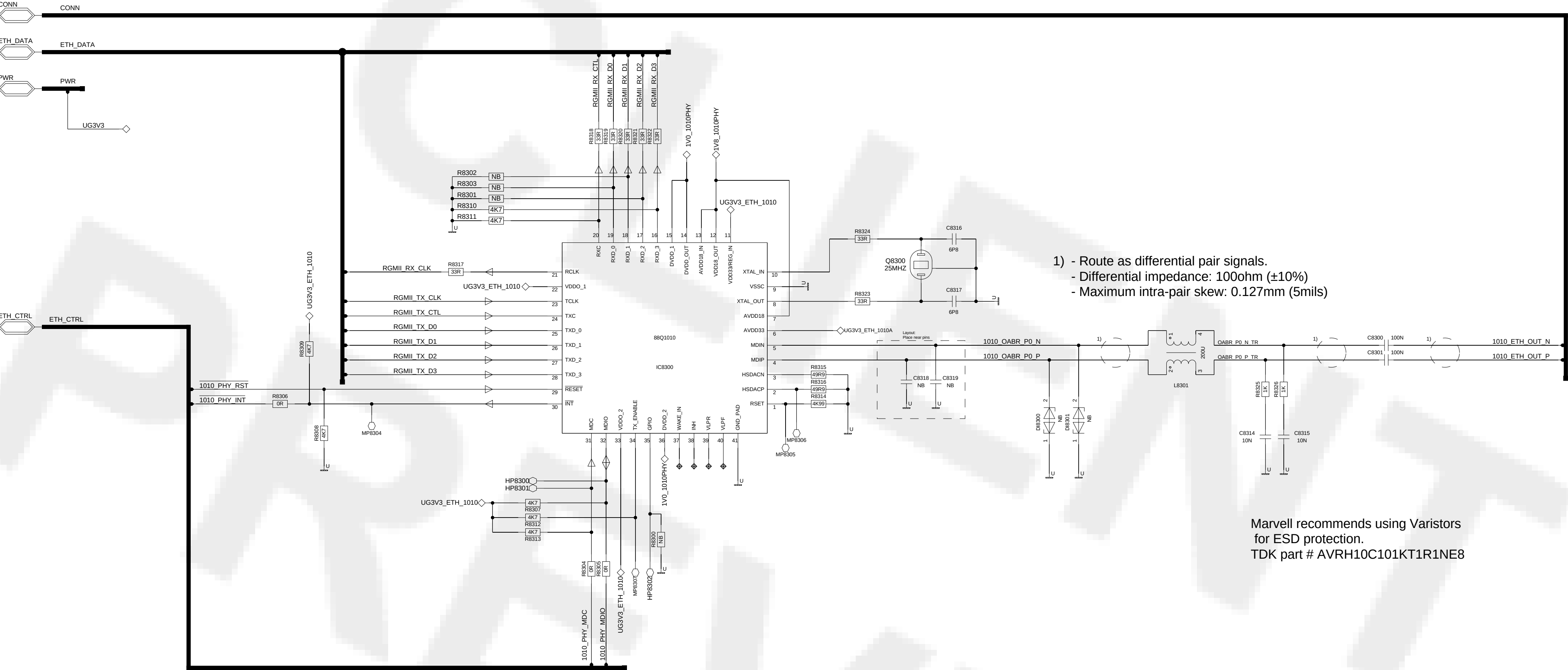
Rev.	type of change		change-no.	date of change	auth. checked
 Copyright reserved Wiring modifications reserved			date		name
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			5	31	
4010_Debug			Type	Variant	Part Number
substitute for:		substitute by:			



Copyright reserved

Writing modifications reserved

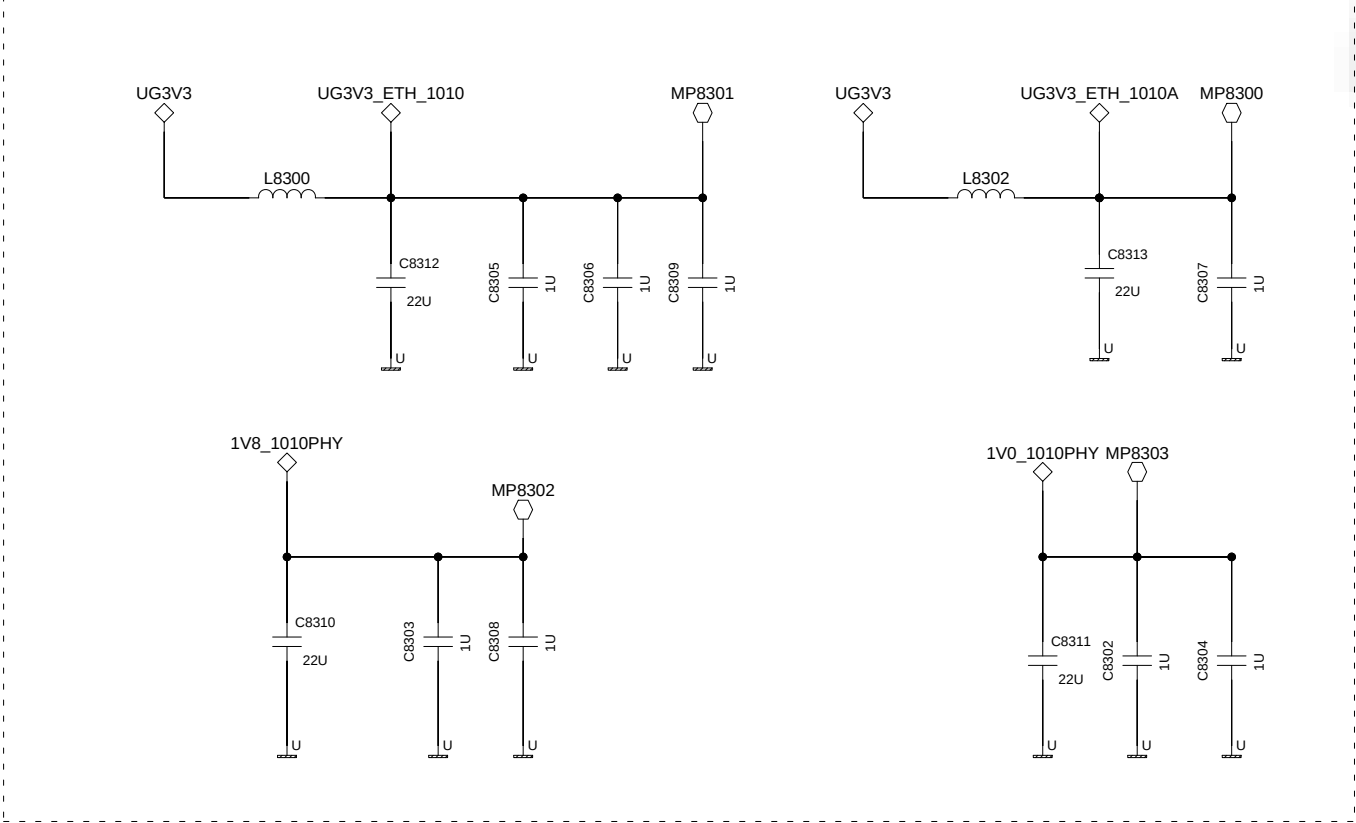
ETH-PHY (2250-2299)



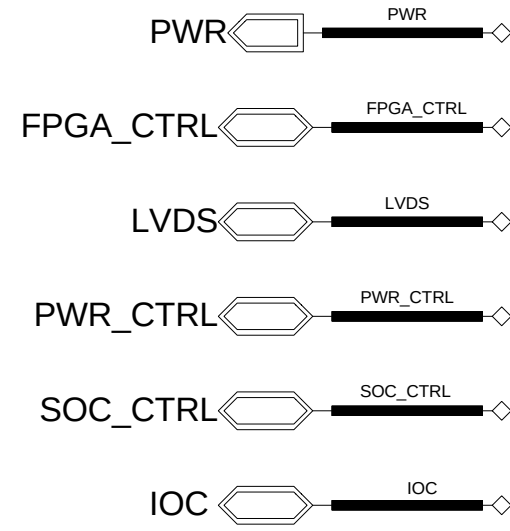
- 1) - Route as differential pair signals.
- Differential impedance: 100ohm (±10%)
- Maximum intra-pair skew: 0.127mm (5mils)

Marvell recommends using Varistors
for ESD protection.
TDK part # AVRH10C101KT1R1NE8

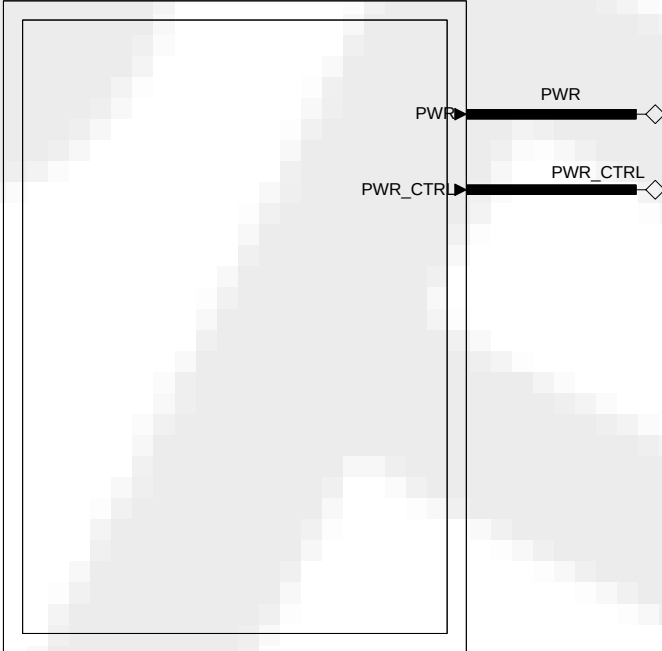
Layout:
Place 1uF caps near each Power Pin



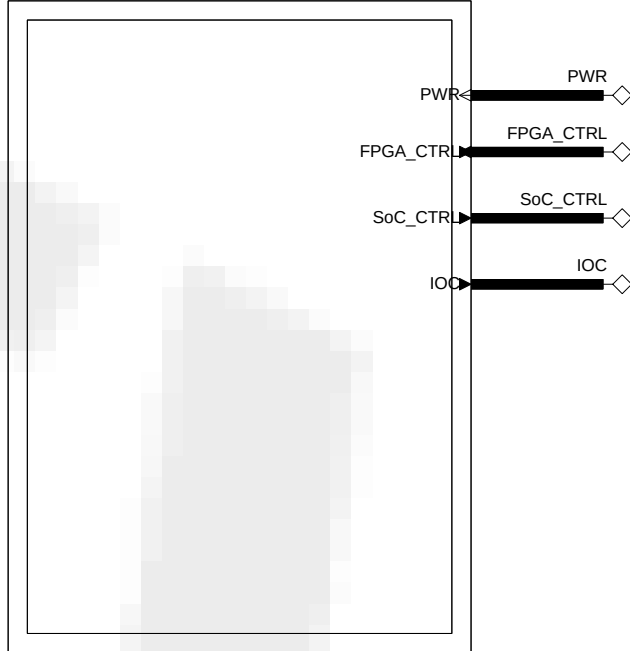
Rev.	type of change					change-no.	date of change	auth.	checked
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4010_ETH_PHY				6	31				
substitute for:				Type	Variant	Part Number			
substitute by:									



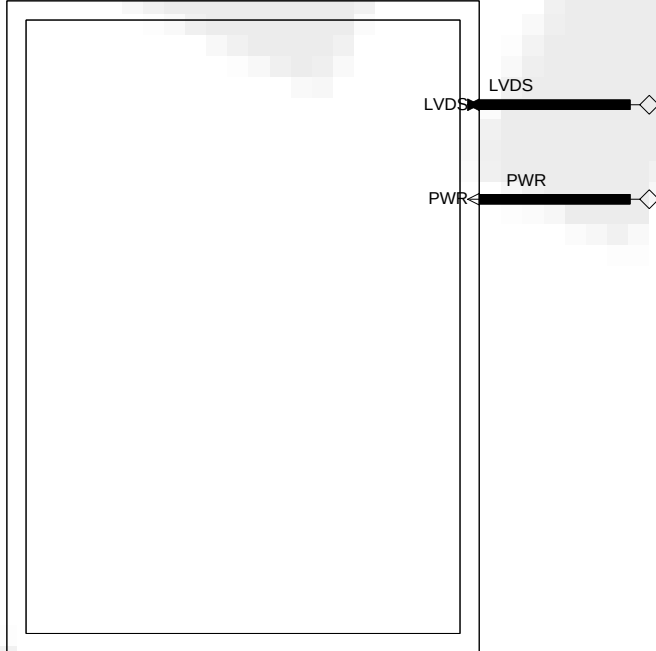
FPGA_POWER



FPGA_CONF

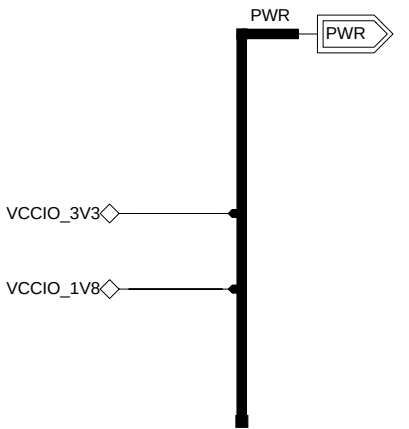
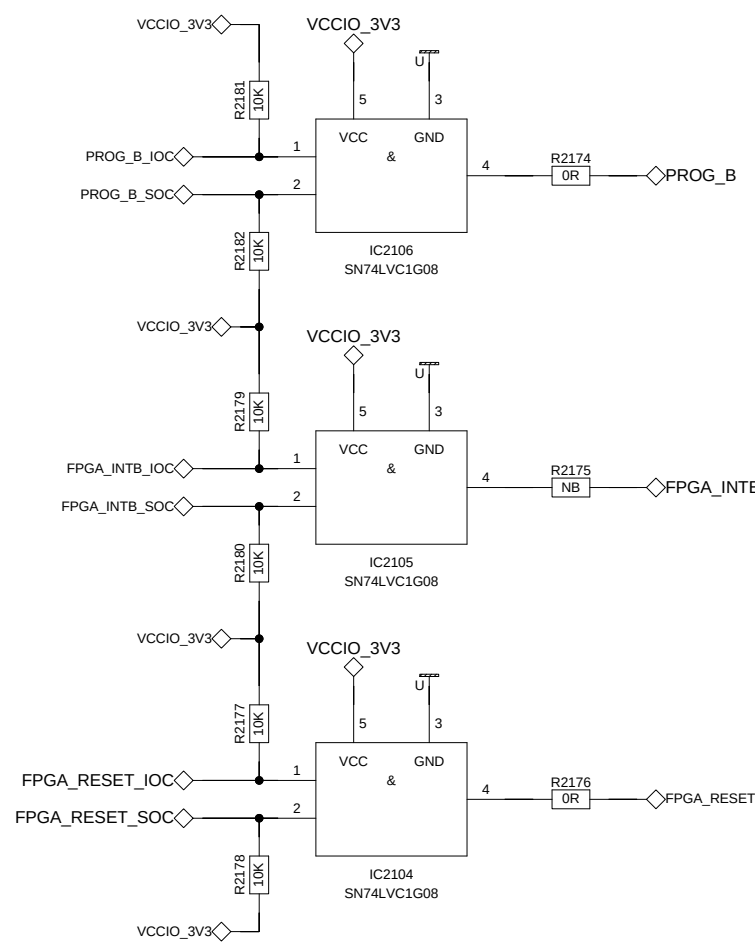
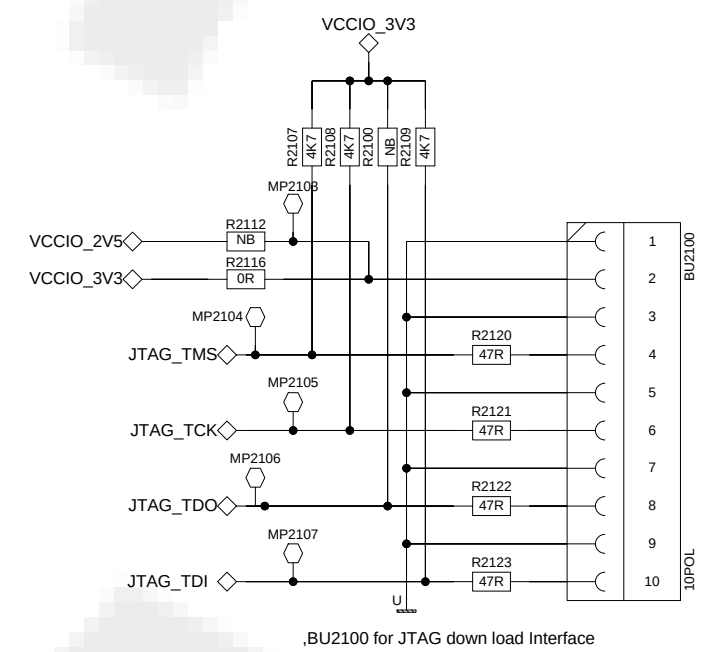
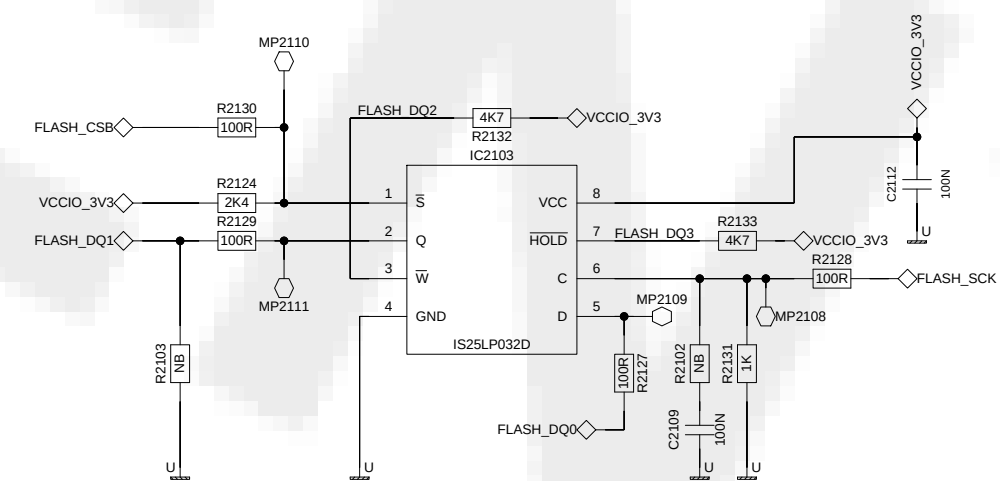
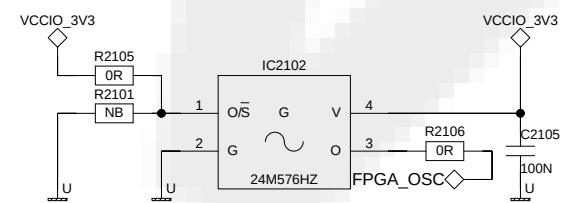
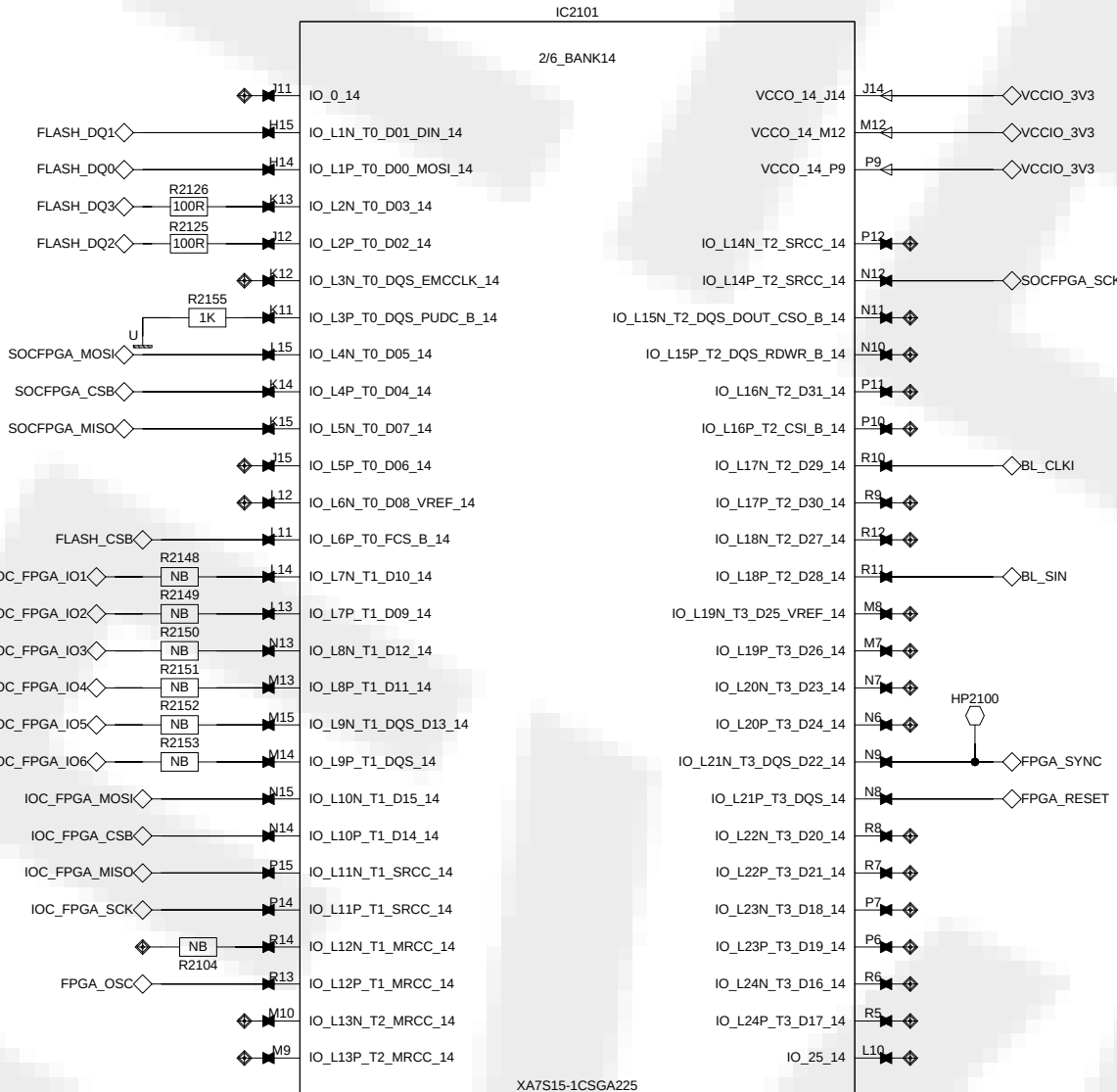
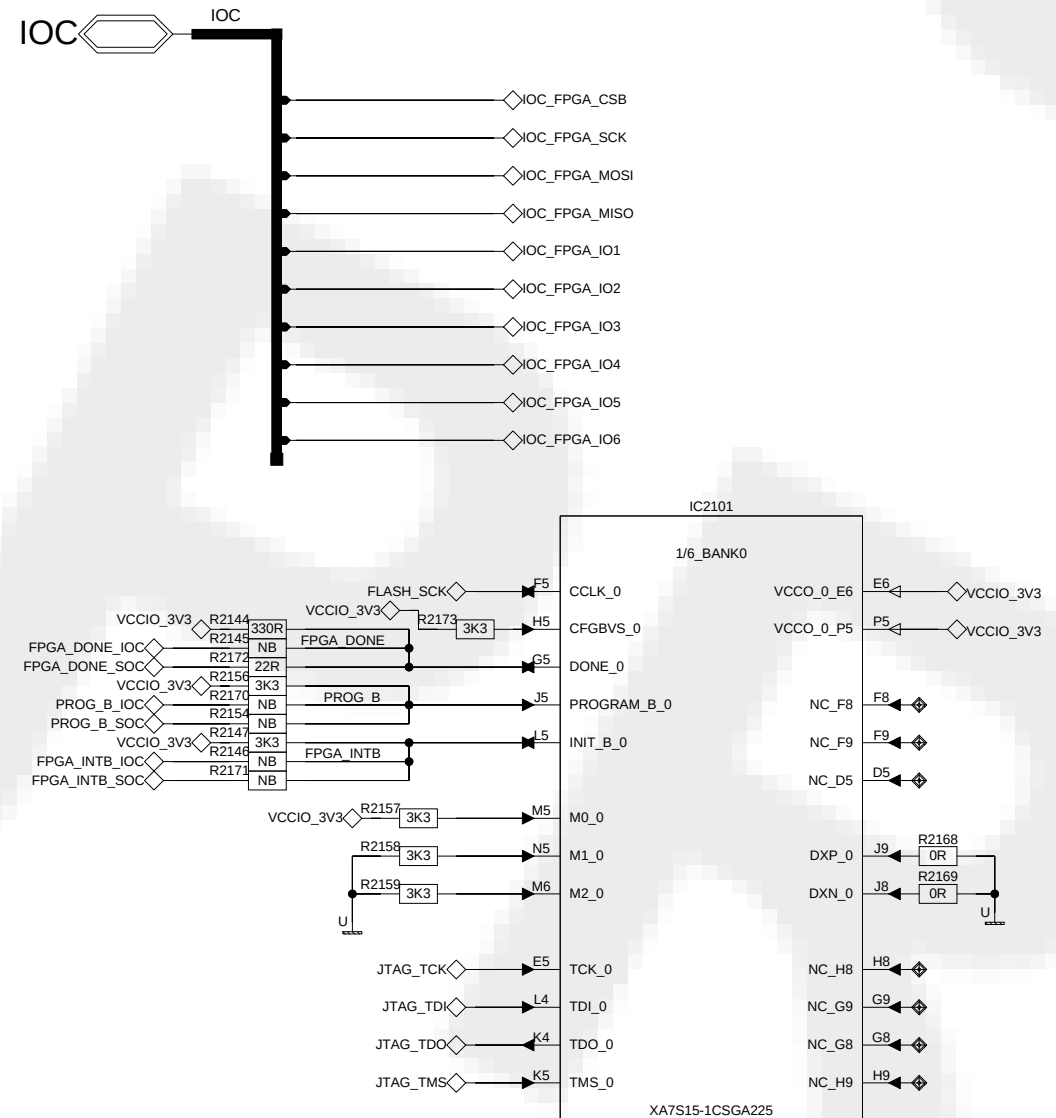
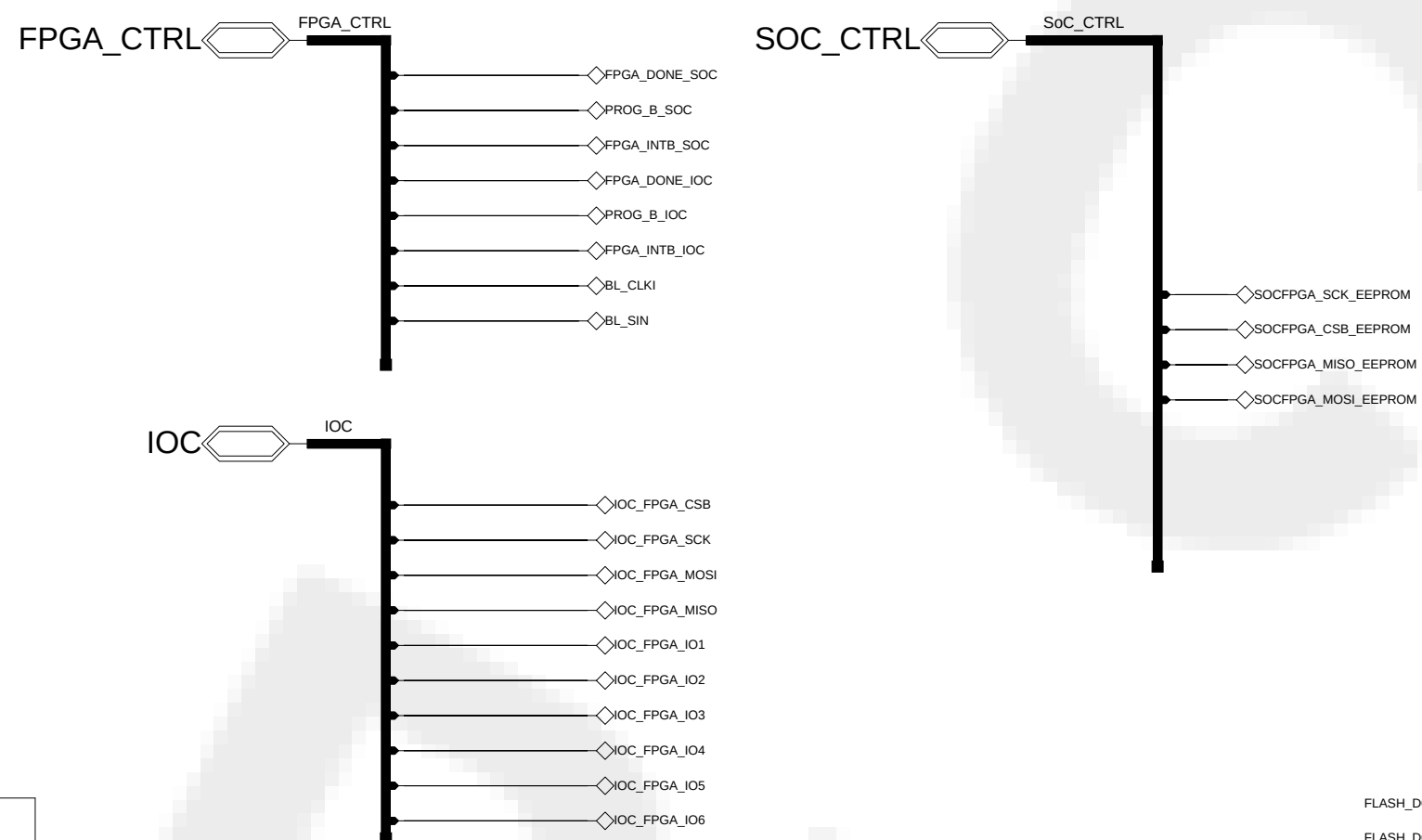


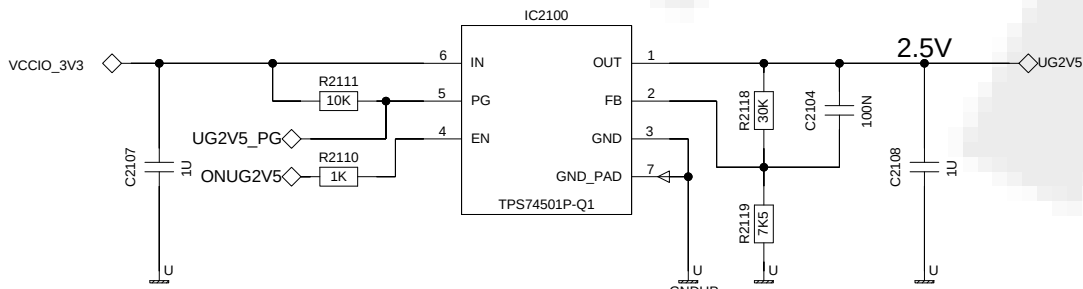
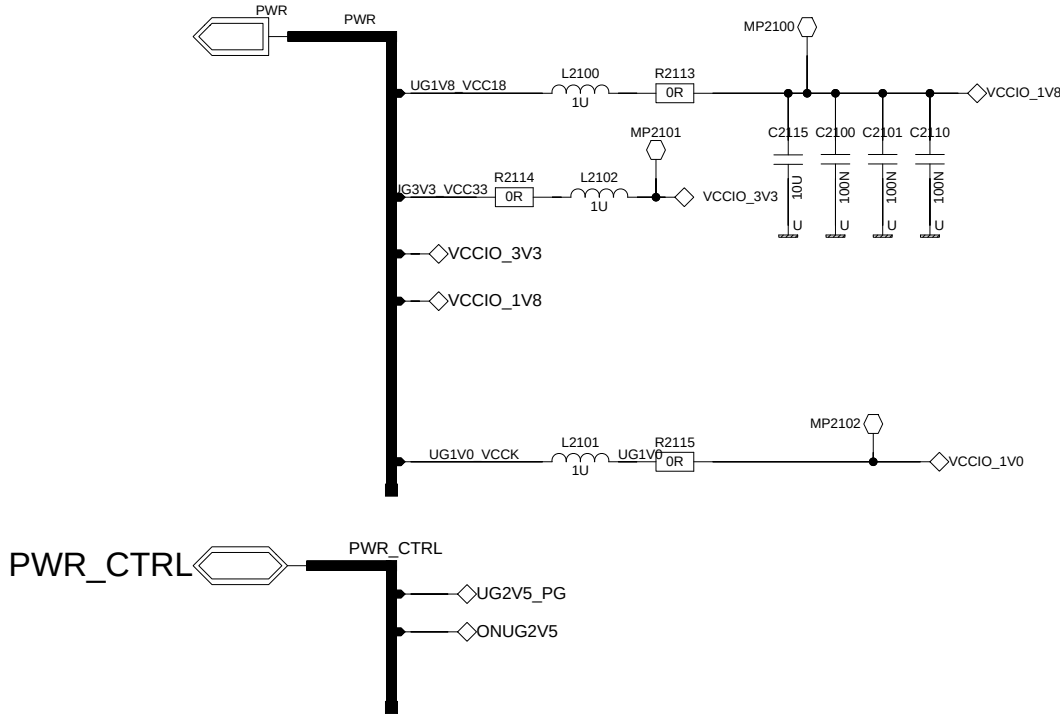
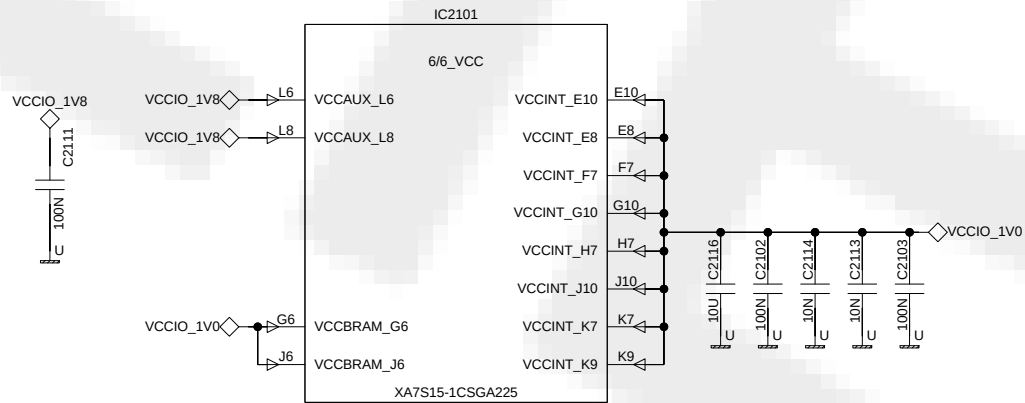
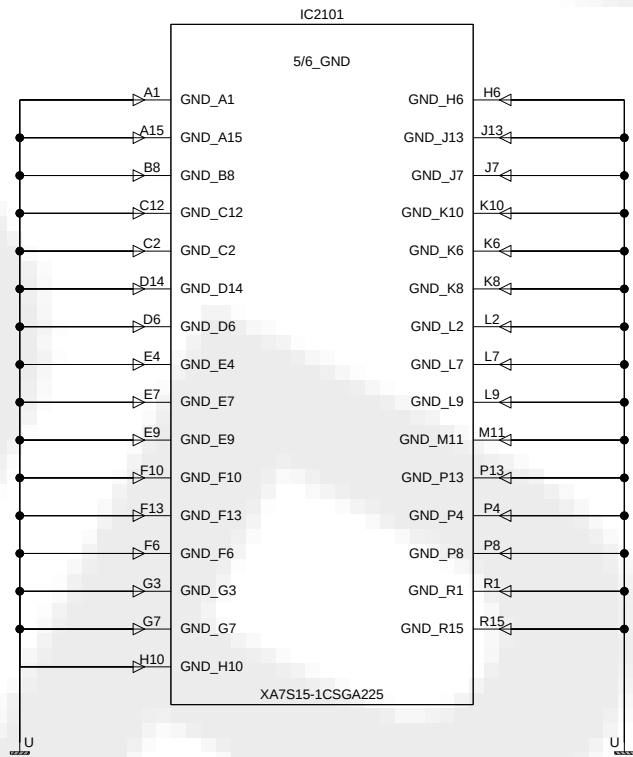
FPGA_VIDEO



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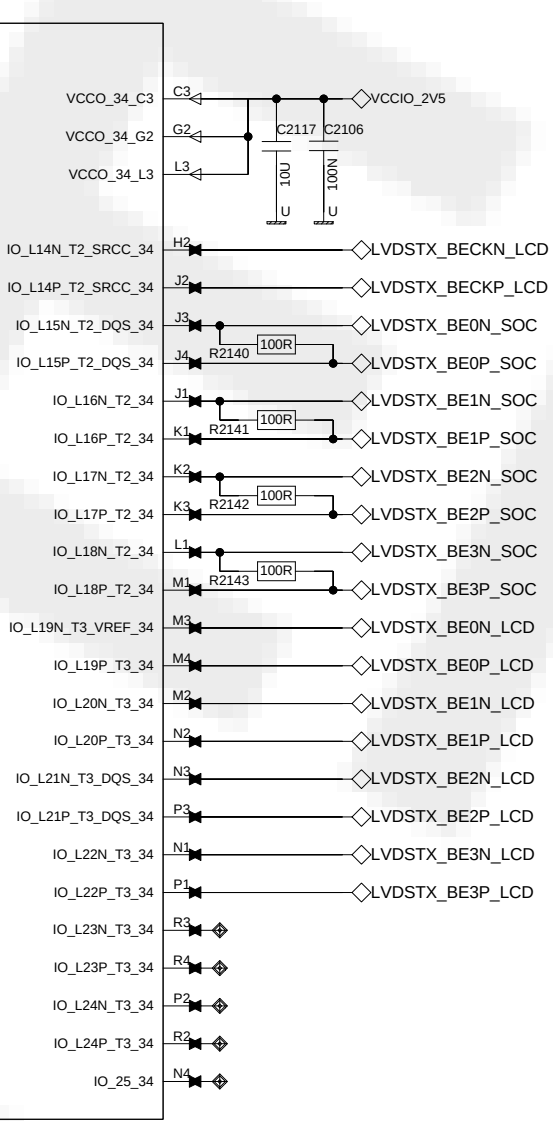
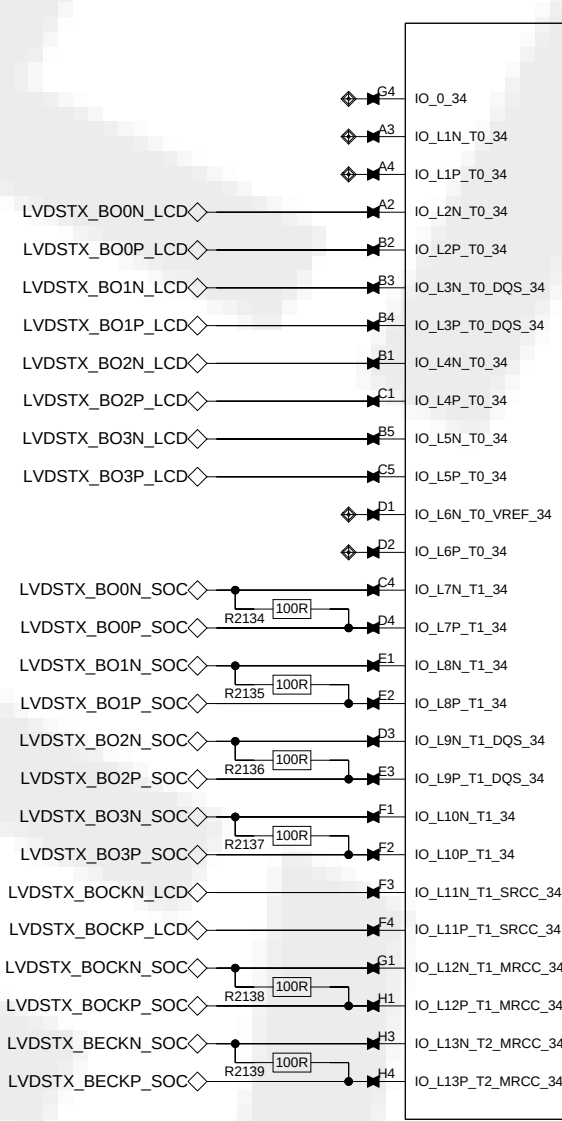
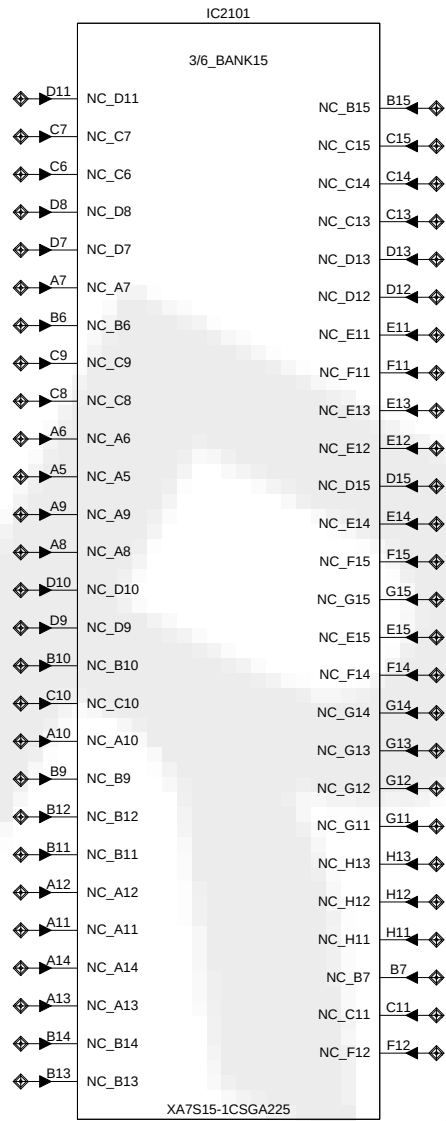
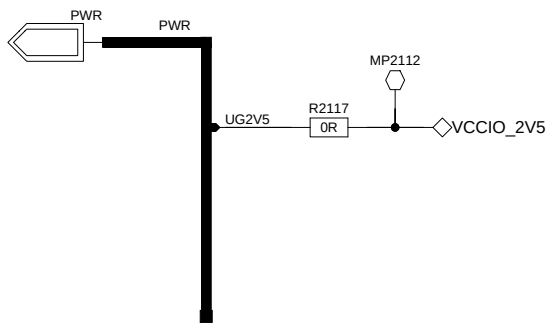
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						Type	Variant	Part Number	
						C143			
substitute for:			substitute by:						





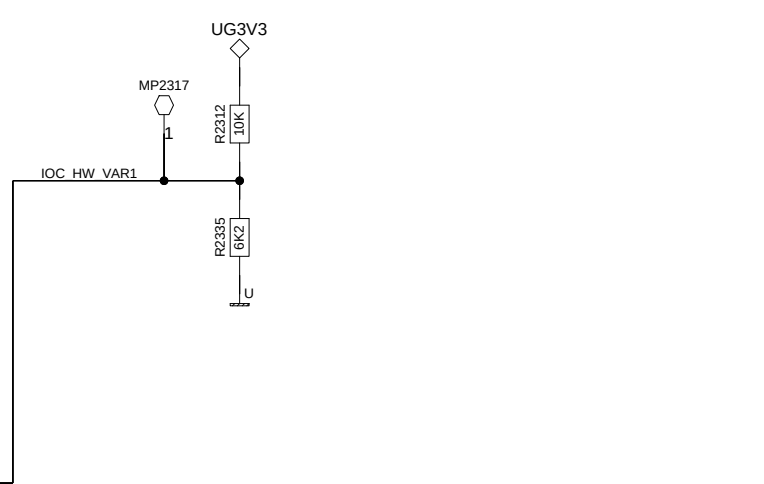
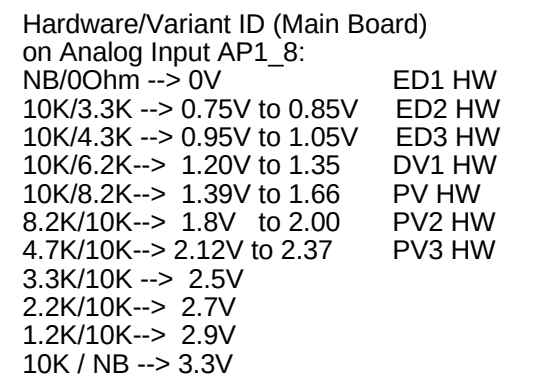
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Sheet		Sheets of		Sample	
9		31			
Type		Variant		Part Number	
substitute for:		substitute by:		C143	



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		Copyright reserved	date		name			
			created					
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FPGA_VIDEO					10	31		
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					C143			

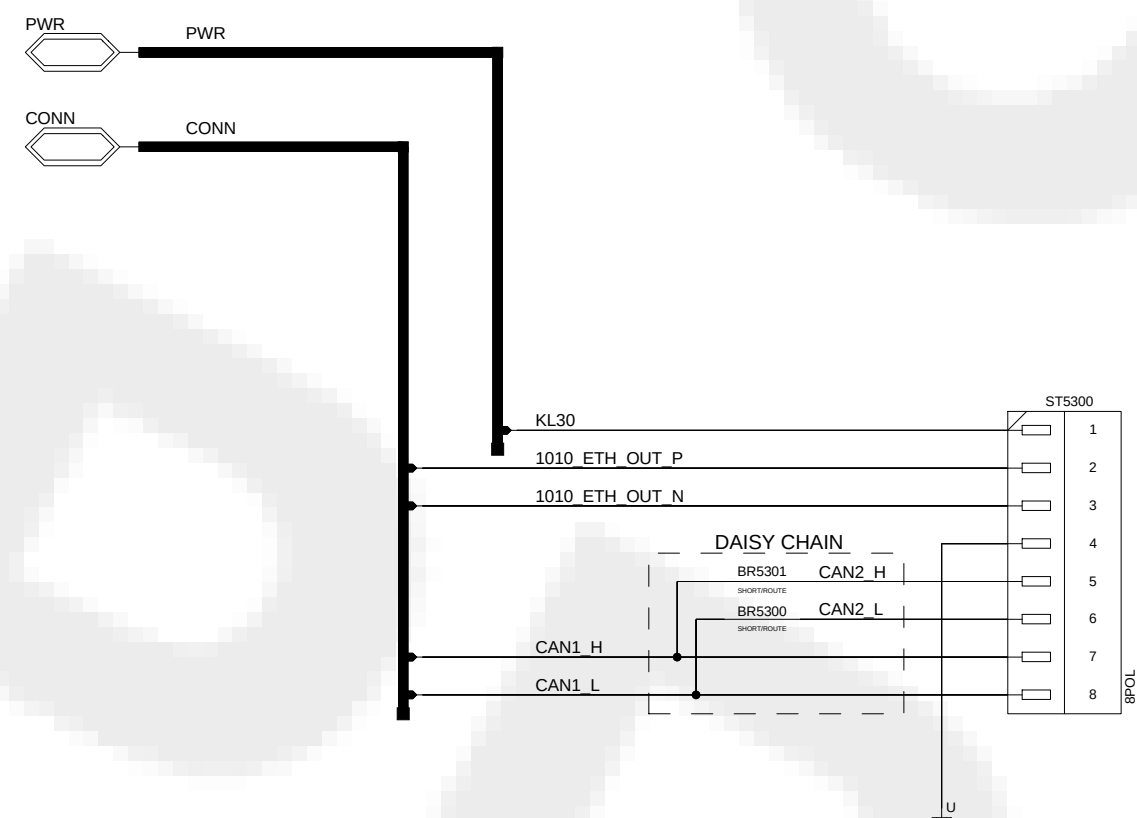
[illegible]

RH850 I2C BUS COMPONENTS	
DEVICE	7-BIT ADDRESS
IOC EEPROM (IC2301)	0x50
PMIC1 PF8100 (IC14200)	0x08

INTP0	AWO	IOC_BKLT_FAULT
INTP1	ISO	
INTP2	ISO	
INTP3	ISO	HS_CAN1_RXD
INTP4	ISO	-----
INTP5	AWO	-----
INTP6	AWO	-----
INTP7	AWO	-----
INTP8	AWO	HS_CAN1_IRQ
INTP9	AWO	
INTP10	ISO	IOC_SOC_IPC_IRQ
INTP11	AWO	-----
INTP12	AWO	-----
INTP13	AWO	-----
INTP14	AWO	DEBUG_RX
INTP15	AWO	

Rev.			type of change			change no.	date of change	auth.	checked
 Copyright reserved Writing modifications reserved							date	name	
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acoustic 4030_IDC substitute for:						Sheet 11	Sheets of 31	Sample	
						Type	Variant C143	Part Number	
substitute by:									

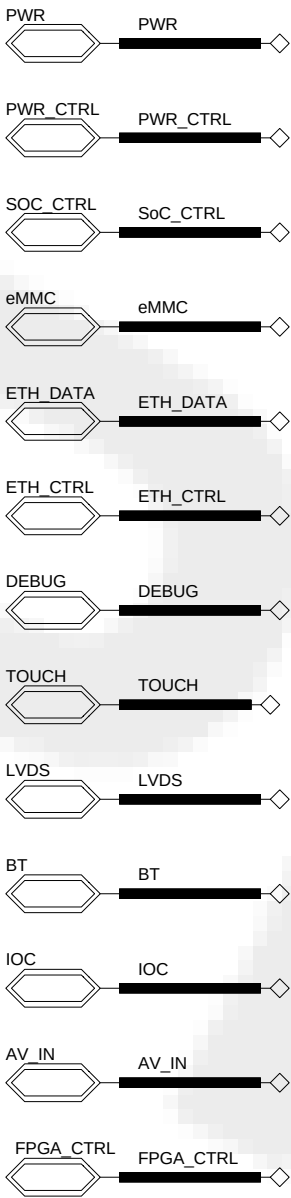
MAIN CONNECTOR (5300-5399)



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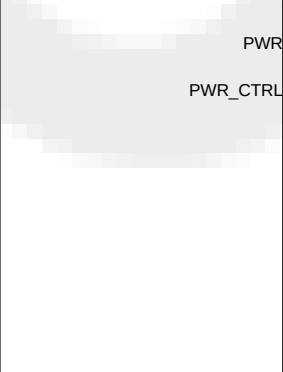
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4010_Main_Connector						Type	Variant		Part Number	
substitute for:			substitute by:							

[illegible]



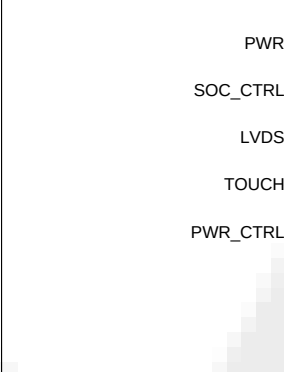
SOC (10000-10999)

4010 SoC Power



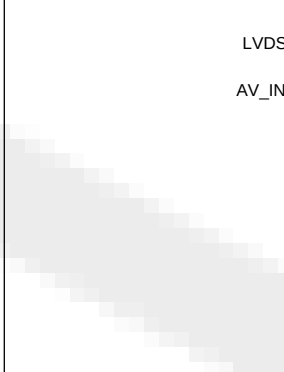
10500 - 10599

4010 SoC_CTRL



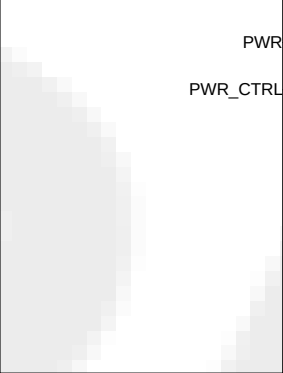
10600 - 10699

4010 SoC Video



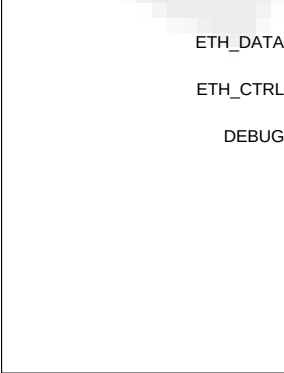
10700 - 10799

4010 SoC LPDDR4



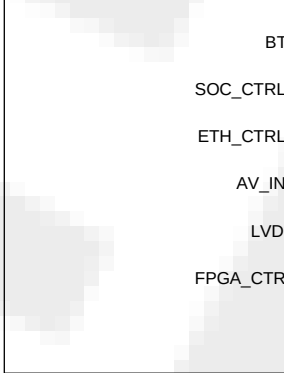
10400 - 10499

4010 SoC ETH PCIe_USB



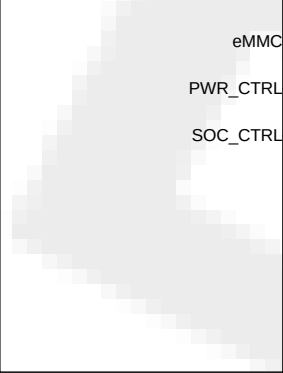
10300 - 10399

4010 SoC Audio



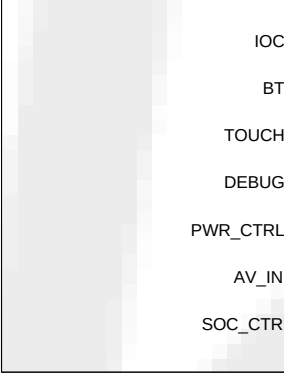
10800 - 10899

4010 SoC eMMC NAND_SD



10100 - 10199

4010 SoC Serial IF



10200 - 10299

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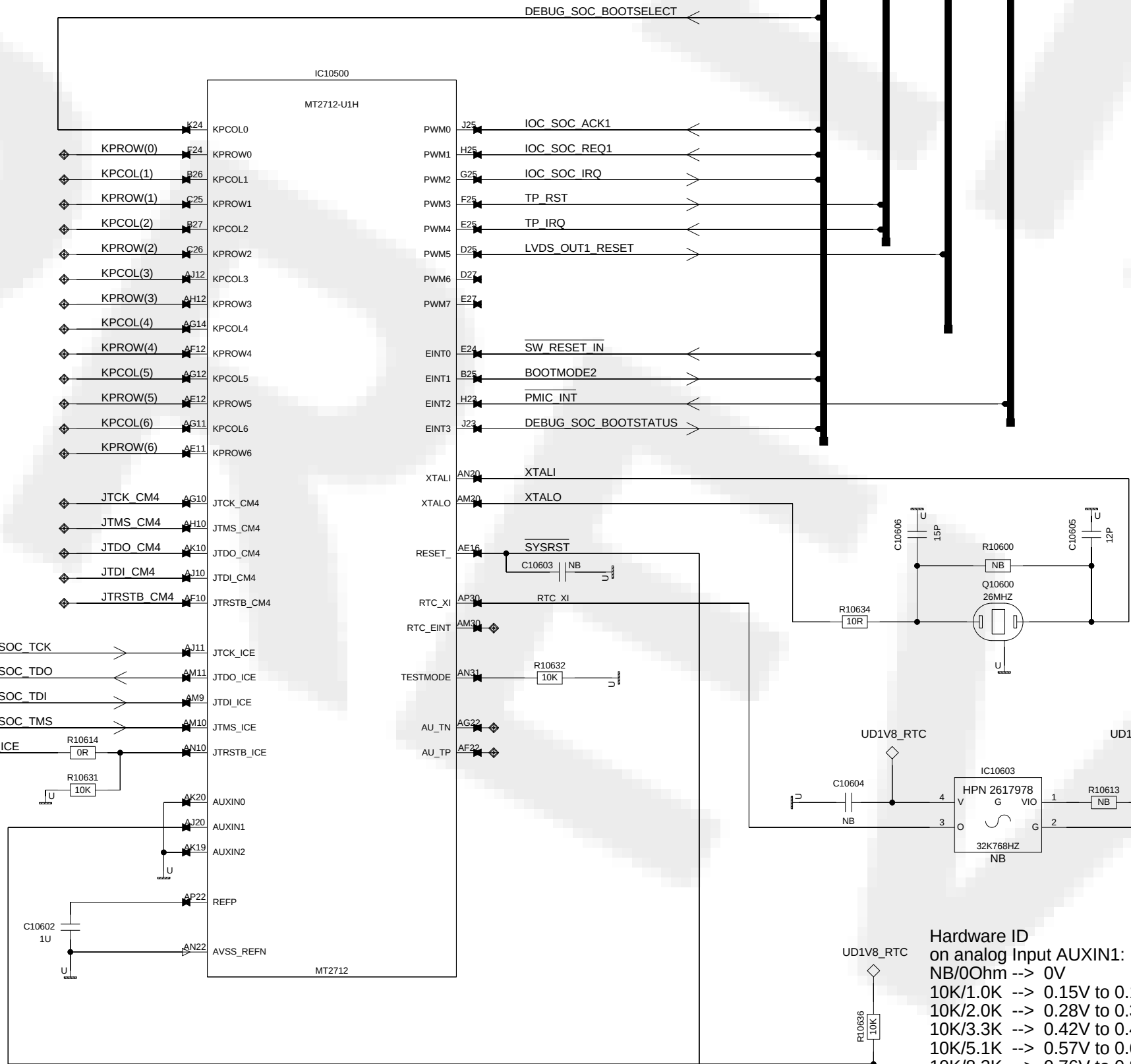
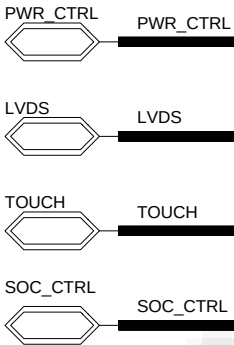
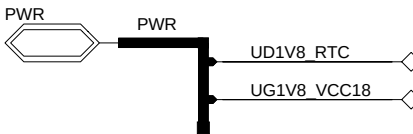
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4010_SoC				15	31					
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substitute by:										

SOC_AUDIO (10800-10899)



Rev.	type of change	change-no.	date of change	auth.	checked
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Sheet		Sheets of		Sample	
4010_Soc_Audio		16		31	
substitute for:		Type		Variant	
		substitute by:		Part Number	

SOC_CTRL (10600-10699)



Hardware ID
on analog Input AUXIN1:
NB/OOhm --> 0V
10K/1.0K --> 0.15V to 0.18V
10K/2.0K --> 0.28V to 0.32V
10K/3.3K --> 0.42V to 0.47V
10K/5.1K --> 0.57V to 0.65V
10K/8.2K --> 0.76V to 0.86V
8.2K/10K --> 0.93V to 1.05V
5.1K/10K --> 1.12V to 1.26V
2.2K/10K --> 1.39V to 1.55V
0Ohm/NB --> 1.71V to 1.89V

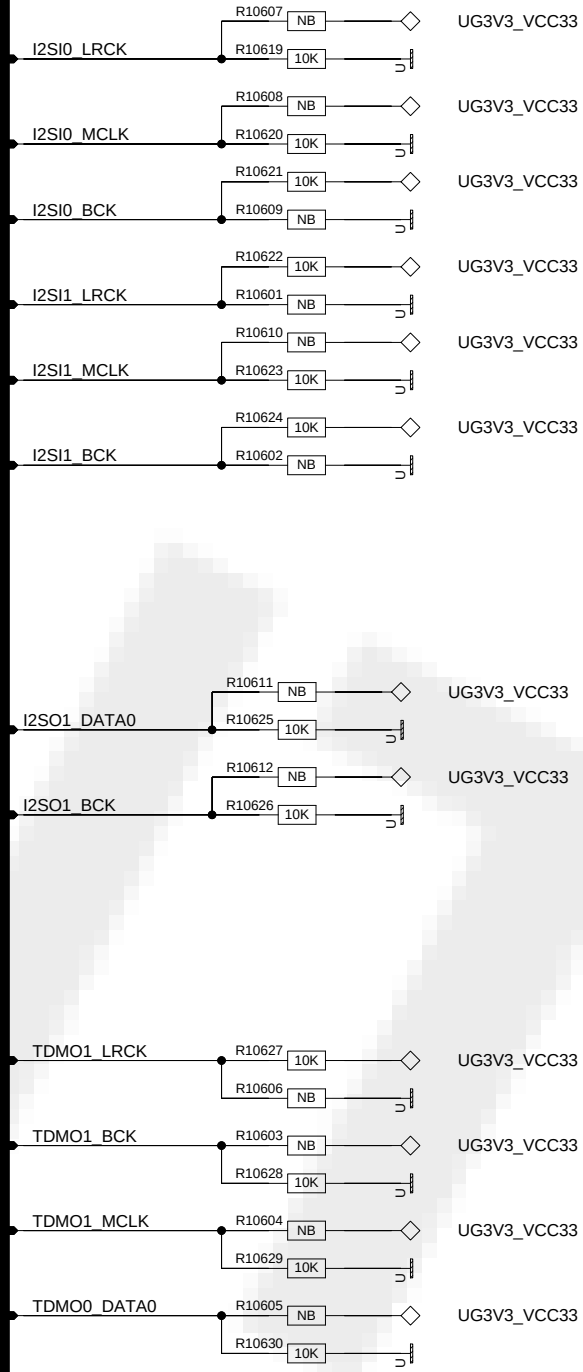
ED1 HW
ED2 HW
ED3 HW
DV HW
PV HW
PV2 HW
PV3 HW

Mode	multi-used		multi-used		multi-used	
	I2SI0_LRCK	I2SI0_MCLK	I2SI0_BCK	I2SI1_BCK	I2SI1_LRCK	I2SI1_MCLK
Combo PHY0 for USB	X	X	X	X	X	1
Combo PHY0 for PCIe	X	X	X	X	X	0
Combo PHY1 for USB	X	X	X	X	1	X
Combo PHY1 for PCIe	X	X	X	X	0	X
PCIIE Port0 for RC	X	X	X	1	X	X
PCIIE Port0 for EP	X	X	X	0	X	X
PCIIE Port1 for RC	X	X	1	X	X	X
PCIIE Port1 for EP	X	X	0	X	X	X
Rom Boot	X	0	X	X	X	X
Nor-flash bypass rom boot	X	1	X	X	X	X
CPU32 default on	0	X	X	X	X	X
CPU64 default on	1	1	X	X	X	X

IPC1POR/Boot Mode					
I2SO1_DATA0 status[1]	L	L	H	H	
I2SO1_BCK status[0]	L	H	L	H	
1	EMMC	NAND	NOR	SD	
2	SD	EMMC	NAND	NOR	
3	NOR	SD	EMMC	NAND	
4	NAND	NOR	SD	EMMC	

Trapping Mode	TDMO1_LRCK	TDMO1_BCK	TDMO1_MCLK	TDMO0_DATA0
Slowest Boot device speed configuration	1	X	X	X
Fastest Boot device speed configuration	0	X	X	X
NAND randomizer: disable	X	0	X	X
NAND randomizer: enable	X	1	X	X
NOR flash address mode: 3-byte	X	X	0	X
NOR flash address mode: 4-byte	X	X	1	X
Alternative boot loader: disable	X	X	X	0
Alternative boot loader: enable	X	X	X	1

Pin Strapping



Combo PHY0 for PCIe
PCIIE Port0 for RC
Rom Boot
CPU32 default on

Rev.	type of change		change-no.	date of change	auth. checked
		Copyright reserved	Wiring modifications reserved	date name	
				created	
				checked	
schematic			Sheet 17	Sheets of 31	Sample
4010_Soc_CTRL			Type	Variant	Part Number
substitute for:		substitute by:			

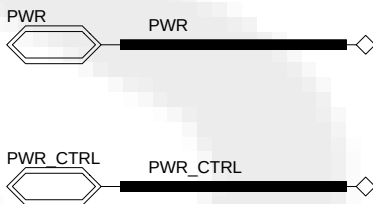
SOC_ETH_PCl_e_USB (10300-10399)



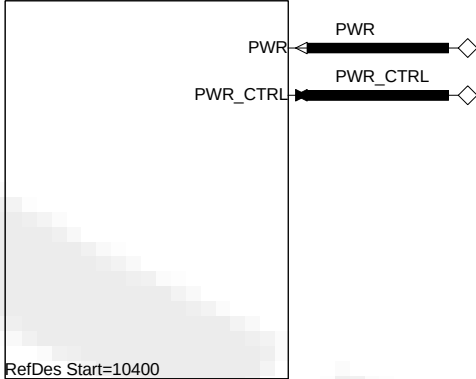
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4010_Soc_ETH_PCIE_USB			Type	Variant	Part Number
substitute for:		substitute by:			

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SOC_LPDDR4 (10400-10499)



4010_SoC_LPDDR4_chA+B



10400 - 10459

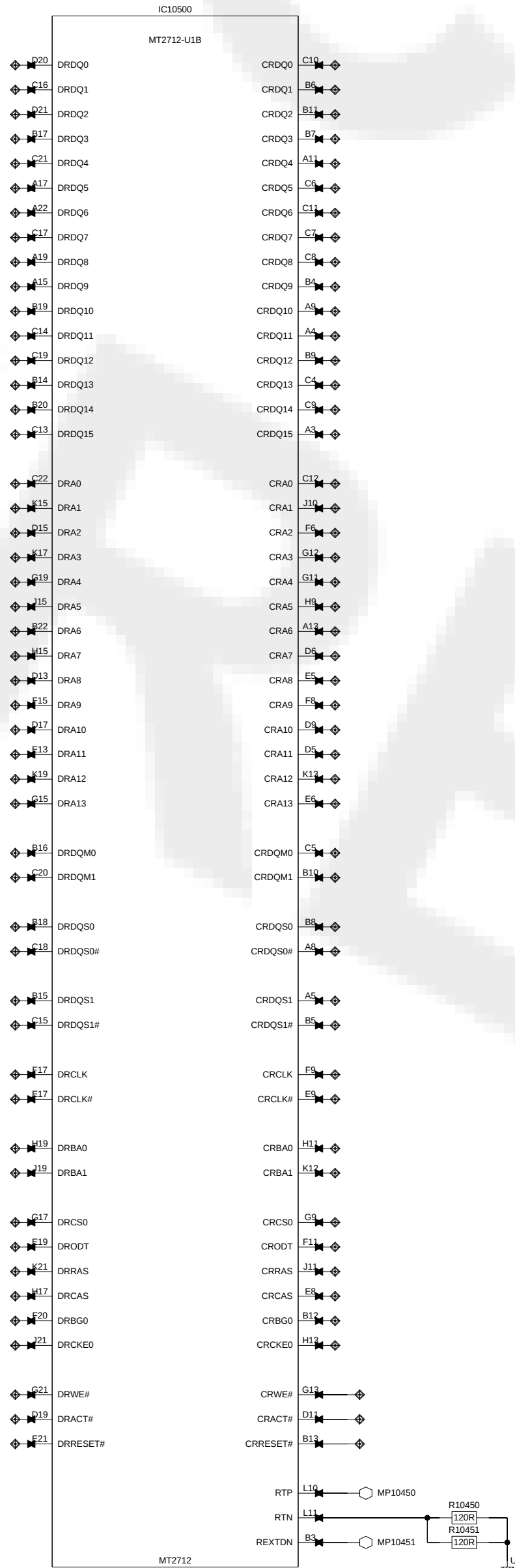
4010_SoC_LPDDR4_chC+D



10450 - 10499

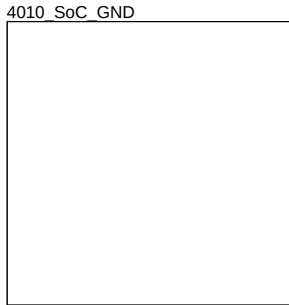
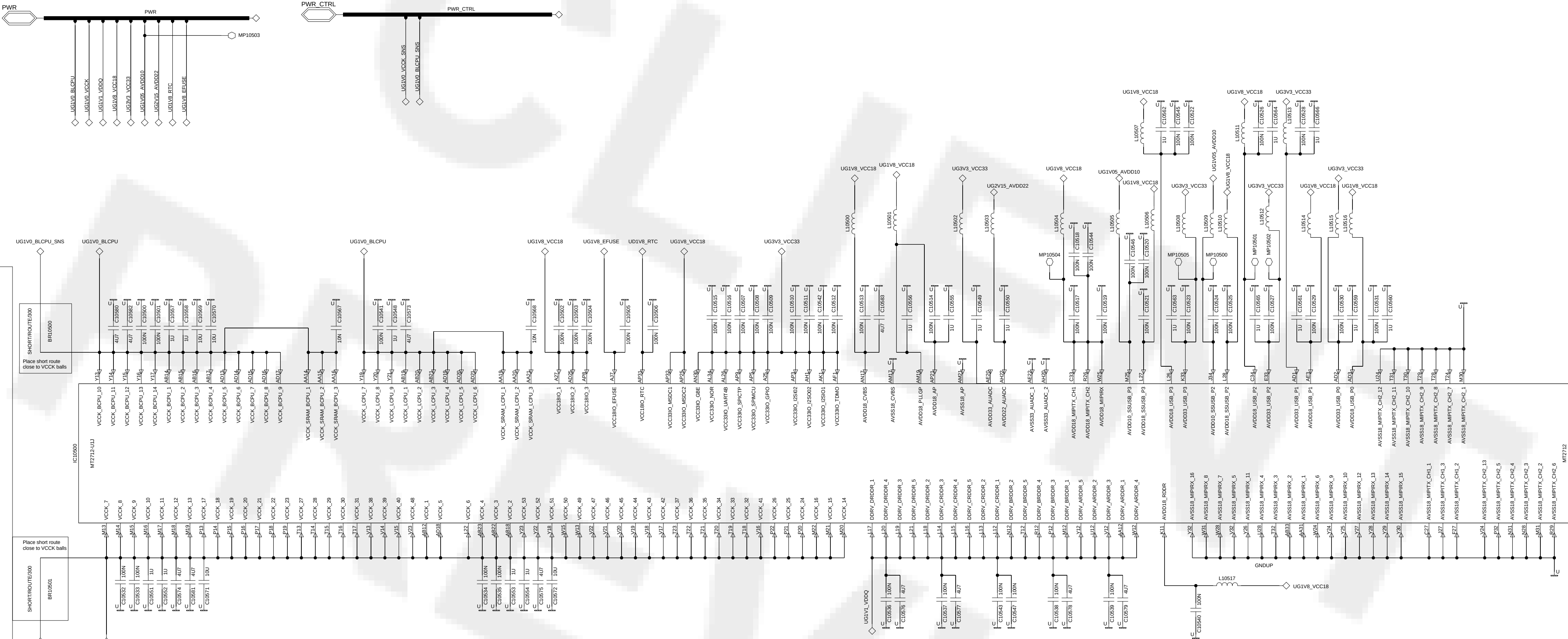
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SOC_LPDDR4 (10450-10499)



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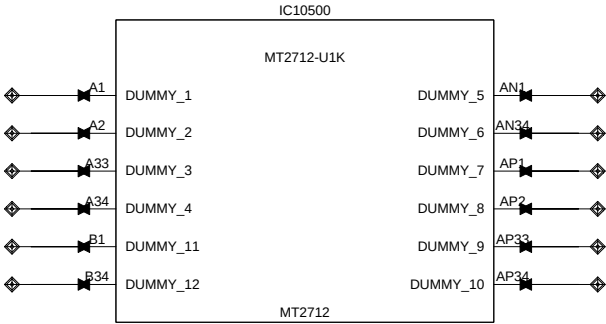
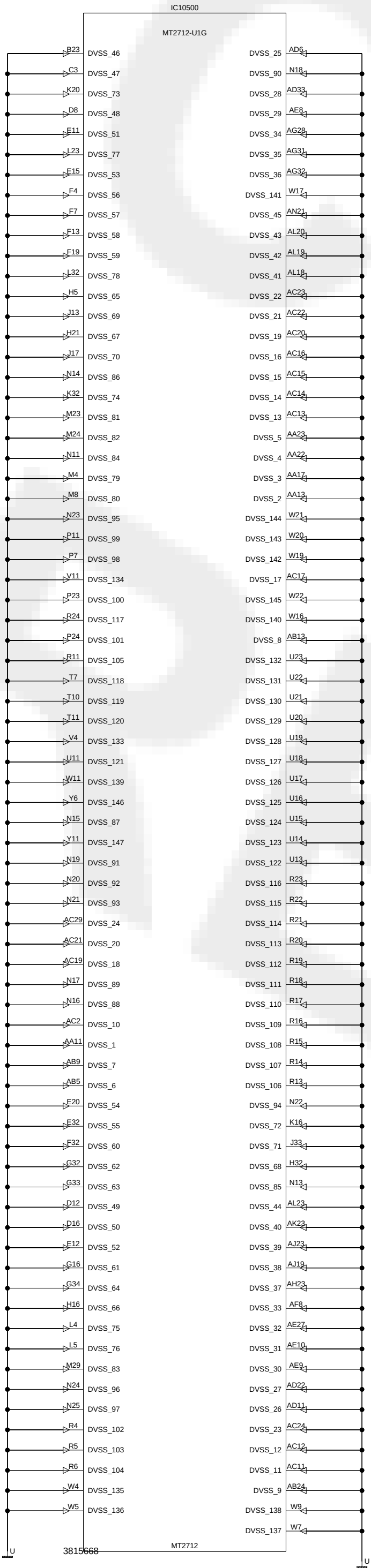
SOC_POWER (10500-10599)



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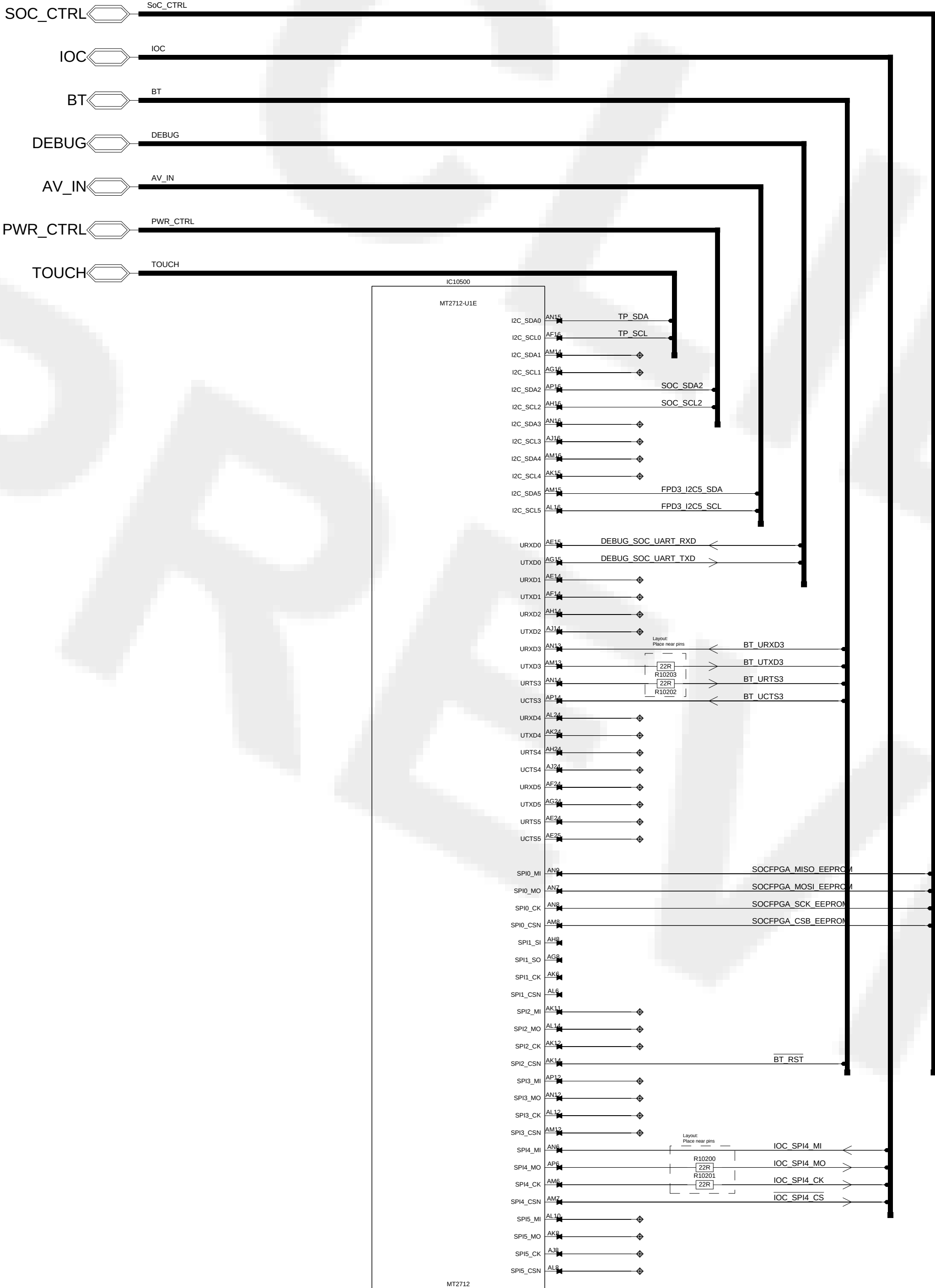
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4010_SoC_Power				22		31				
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schematic					Sheet	Sheets of		Sample			
4010_SoC_GND					23	31					
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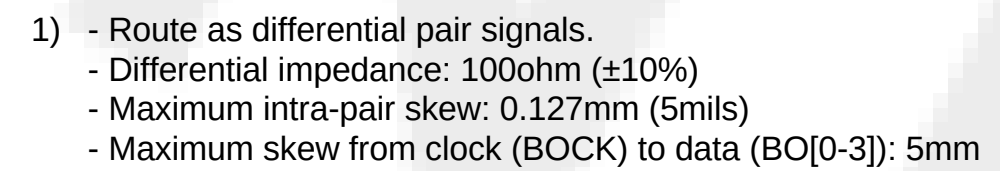
SOC_SERIAL_IF (10200-10299)



Rev.	type of change					change-no.	date of change		auth.	checked
			Copyright reserved	Writing modifications reserved		date		name		
					created					
					checked					
 4010_SoC_Serial_IF						Sheet 24	Sheets of 31		Sample	
						Type	Variant		Part Number	
substitute for:				substitute by:						

LVDS

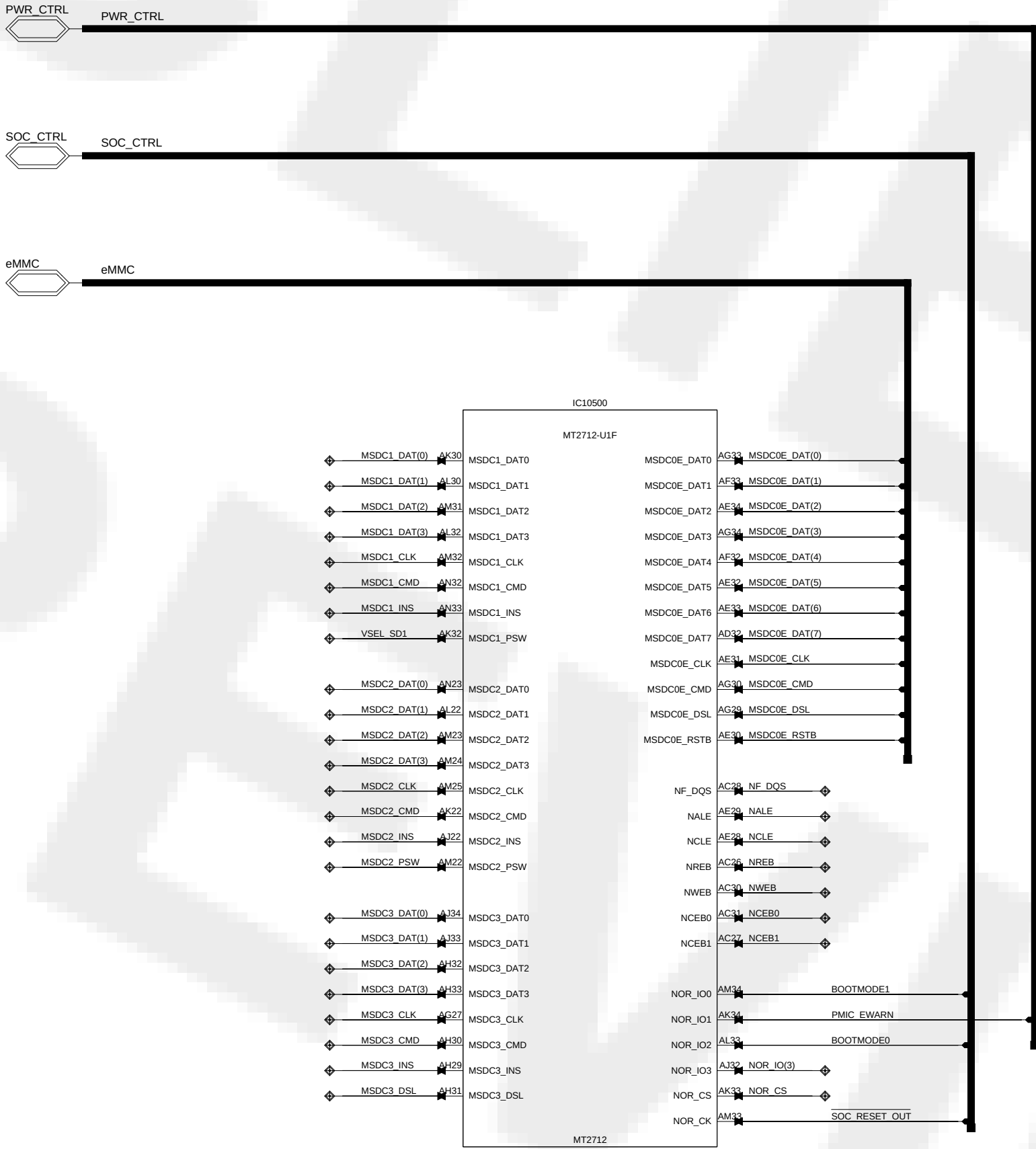
AV IN



- 2) - Route as differential pair signals.
- Differential impedance: 100ohm ($\pm 10\%$)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum skew from clock (BECK) to data (BE[0-3]): 5mm

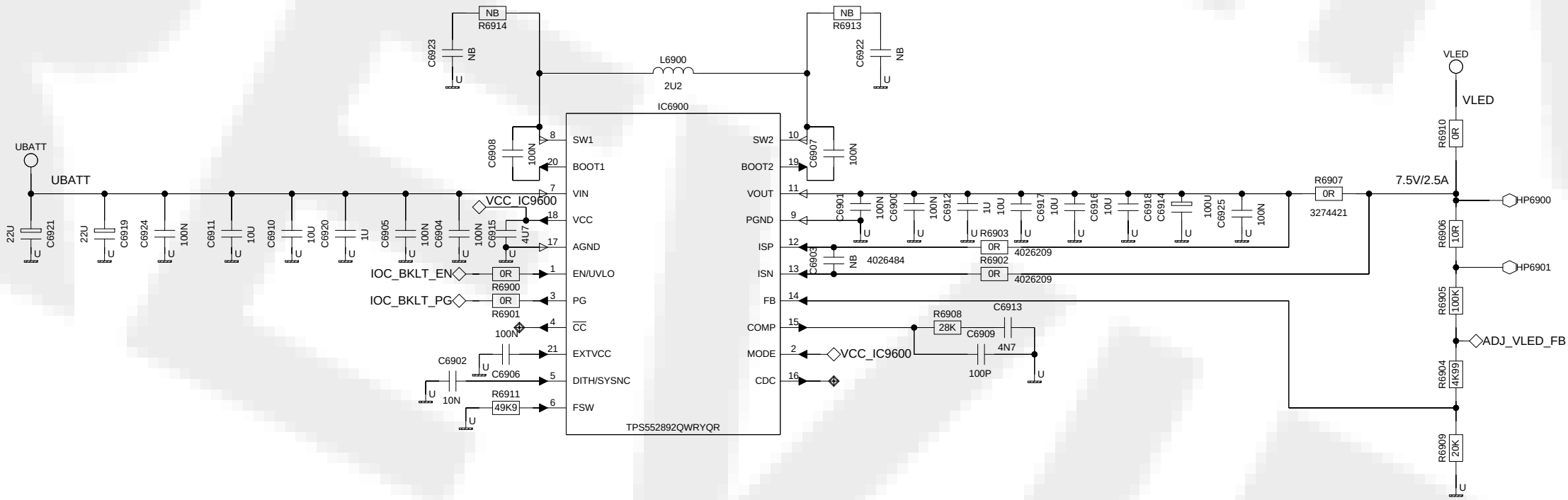
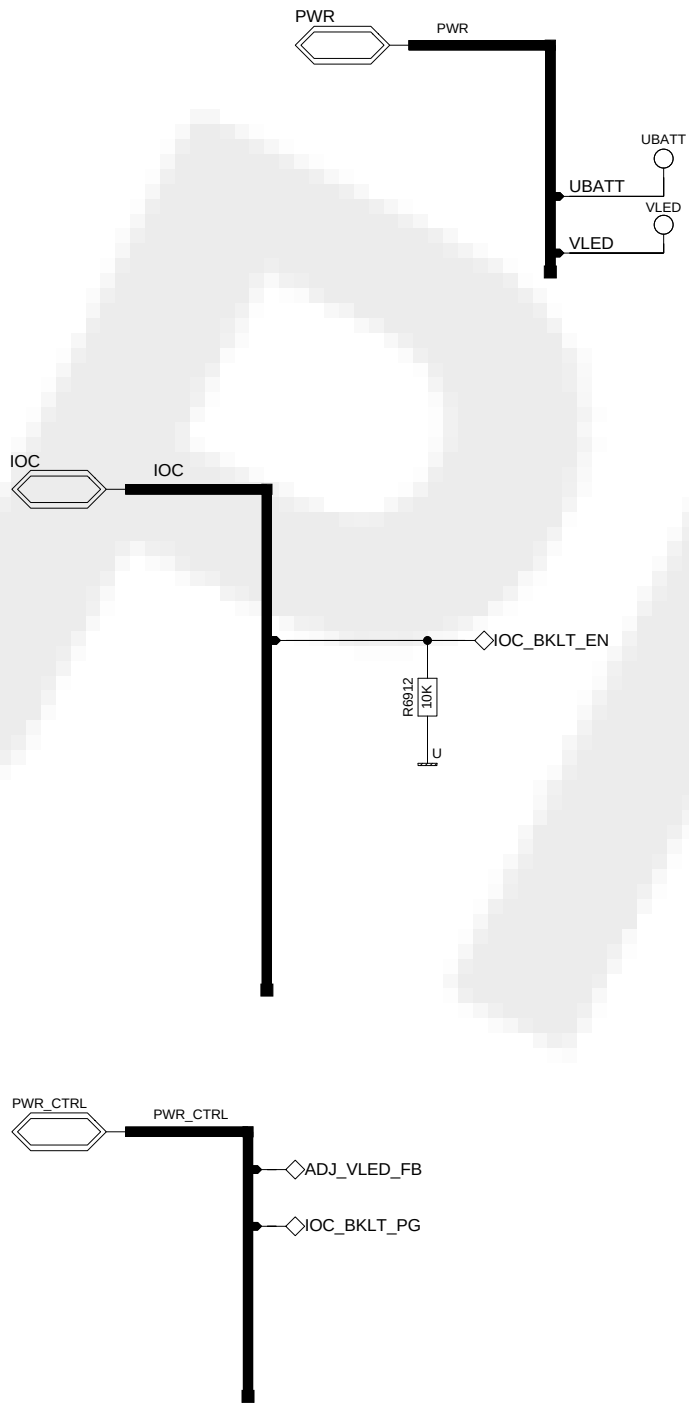
Rev.	type of change					change-no.	date of change		auth.	checked
		Copyright reserved		Wiring modifications reserved		date		name		
						created				
						checked				
 4010_SoC_Video						Sheet 25	Sheets of 31		Sample	
						Type	Variant		Part Number	
substitute for:				substitute by:						

SOC_eMMC_NAND_SID (10100-10199)



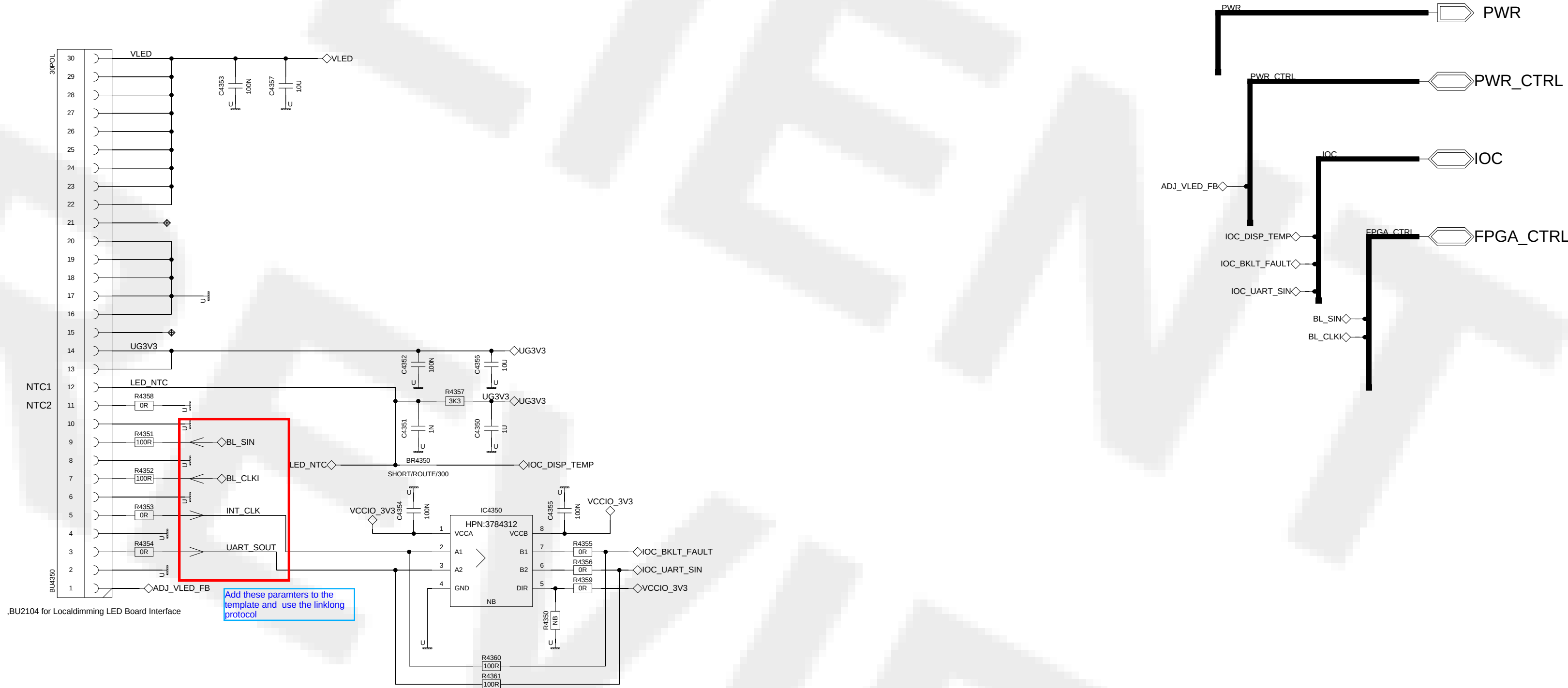
Rev.	type of change					change-no.	date of change		auth.	checked		
			Copyright reserved		Writing modifications reserved		date		name			
							created					
							checked					
schematic							Sheet	Sheets of	Sample			
4010_SoC_eMMC_NAND_SD							26	31				
substitute for:			substitute by:				Type	Variant	Part Number			

TFT BL (6900-6999)

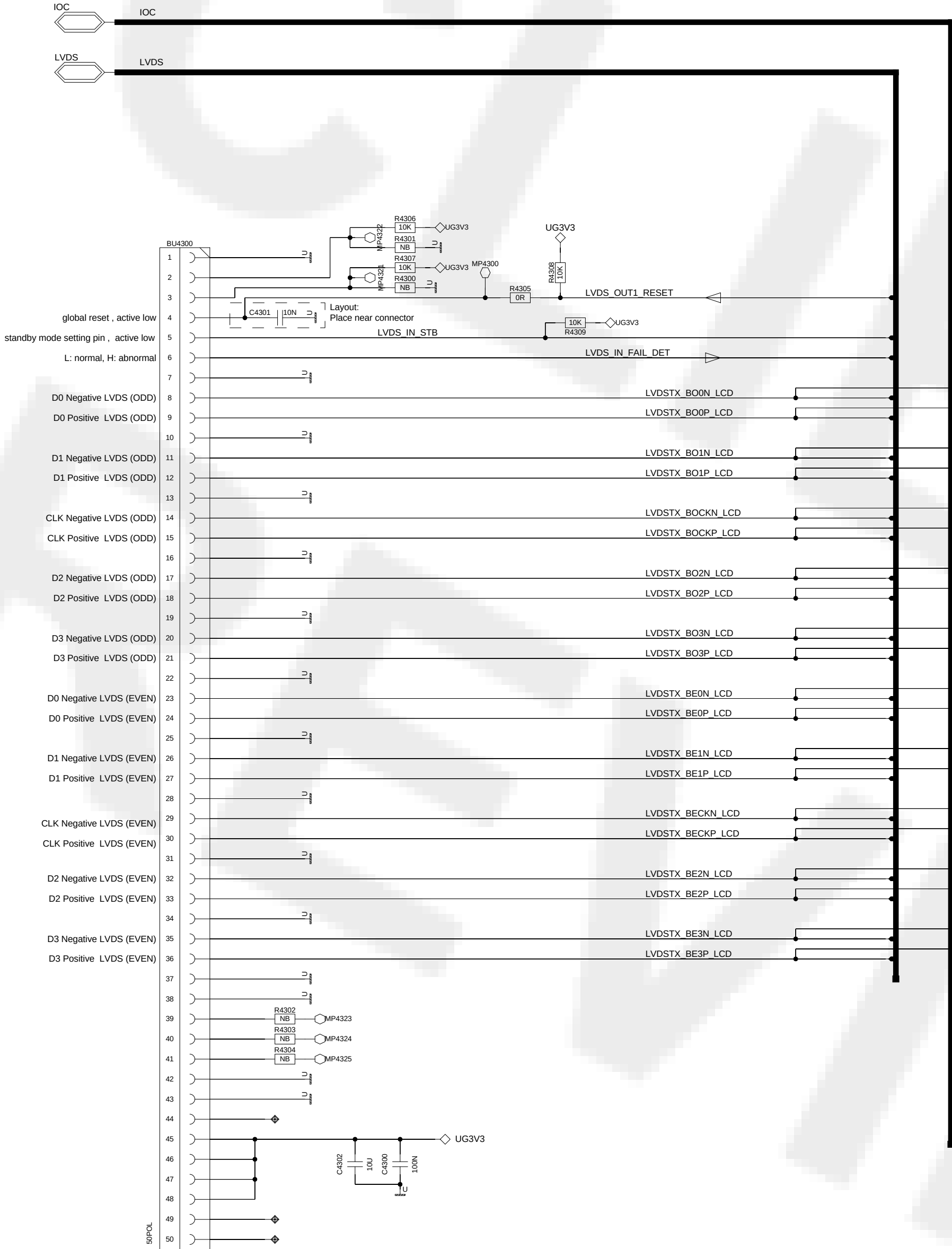
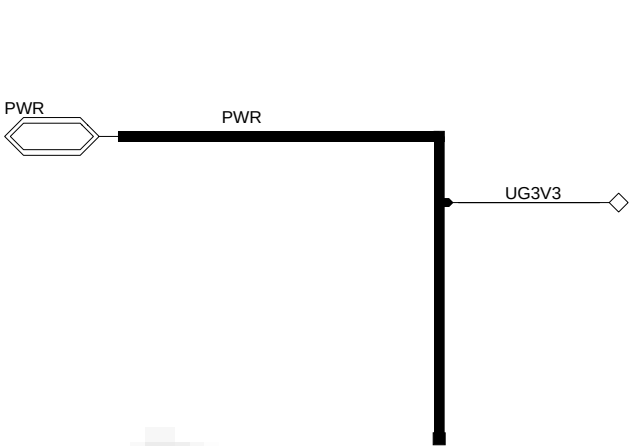


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		Copyright reserved	Wiring modifications reserved	date		name				
				created						
				checked						
schematic				Sheet	Sheets of		Sample			
4010_TFT_BL				27	31					
substitute for:			substitute by:	Type	Variant		Part Number			
					C143					

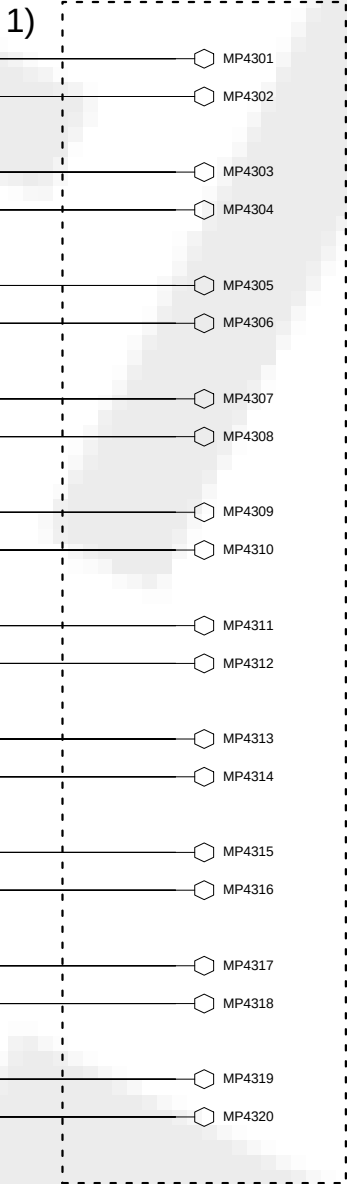
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Rev.	type of change	change-no.	date of change	auth.	checked
			date	name	
		created			
		checked			
		Sheet	Sheets of	Sample	
		28	31		
		Type	Variant	Part Number	
		substitute for:	substitute by:	C143	



Verify pin out during layout



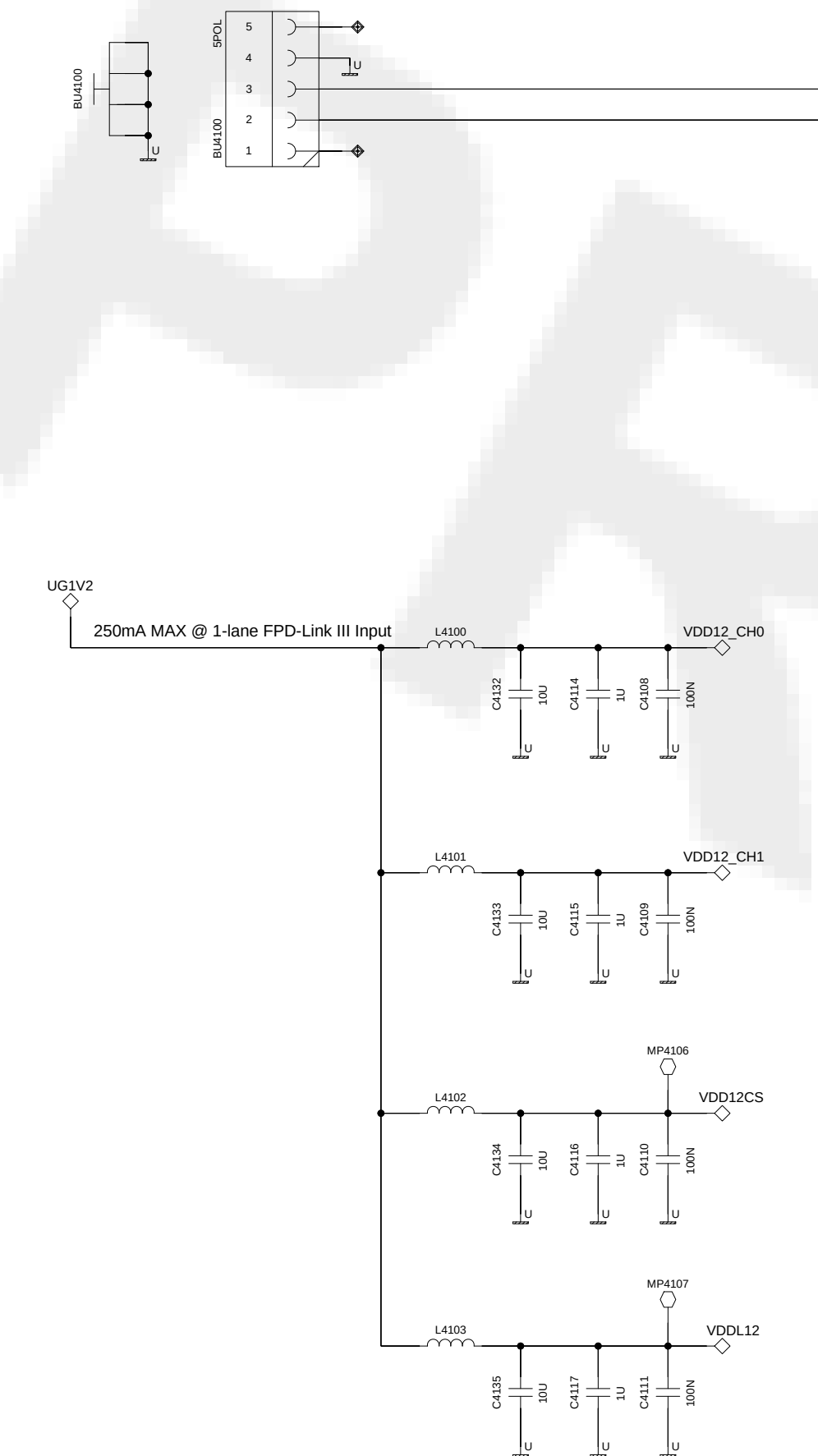
1) Testpoints should be placed as close as possible to an existing VIA

Rev.	type of change		change-no.	date of change	auth. checked
 Copyright reserved Wiring modifications reserved			date		name
			created		
			checked		
schematic			Sheet 29	Sheets of 31	Sample
4010_TFT_Conn			Type	Variant C143	Part Number
substitute for:		substitute by:			

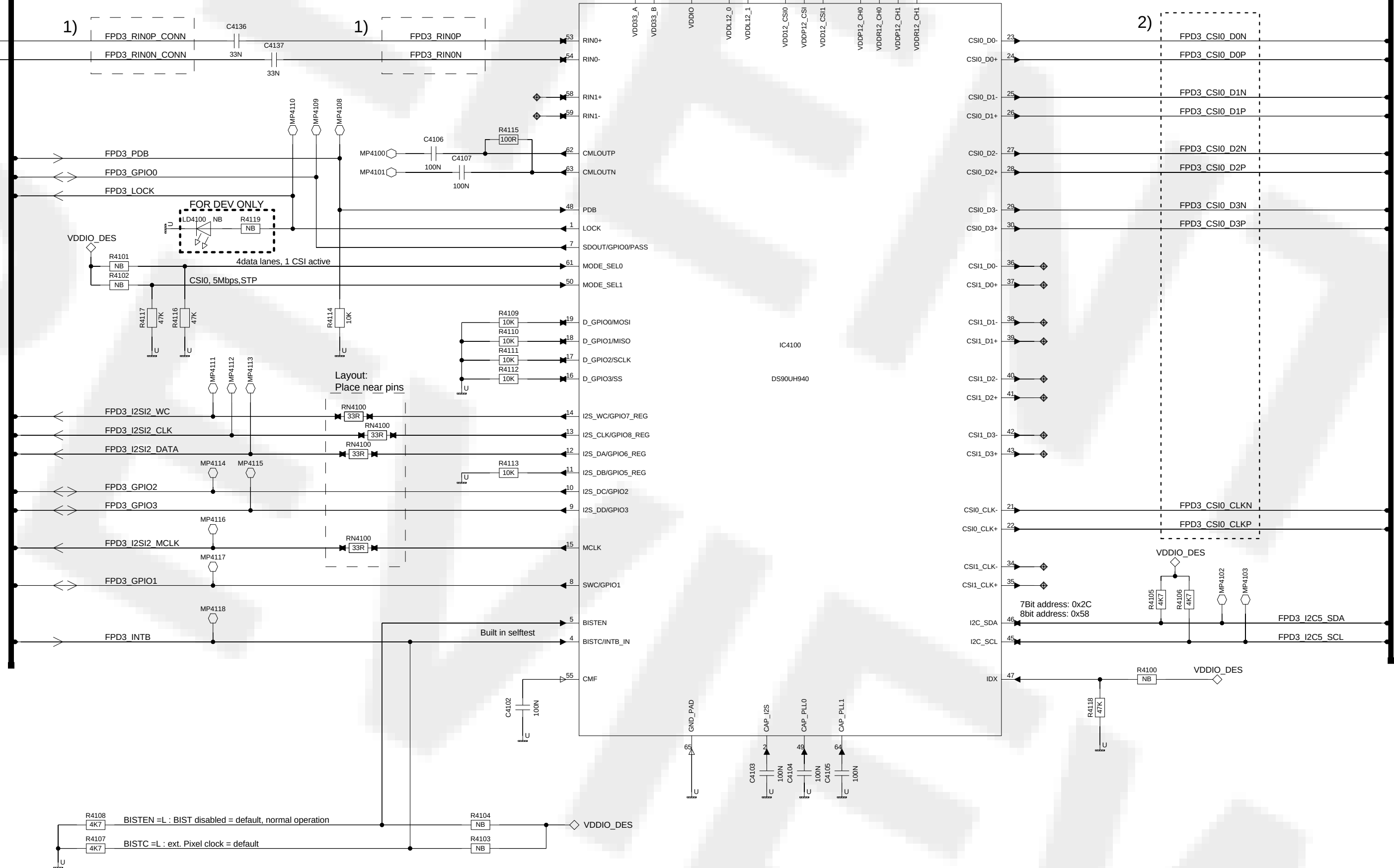


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- 2) - Route as differential pair signals.
- Differential impedance: 100ohm ($\pm 10\%$)
- Maximum intra-pair skew: 0.127mm (5mils)
- Maximum inter-pair skew: 1.27mm (50mils)

[illegible]

